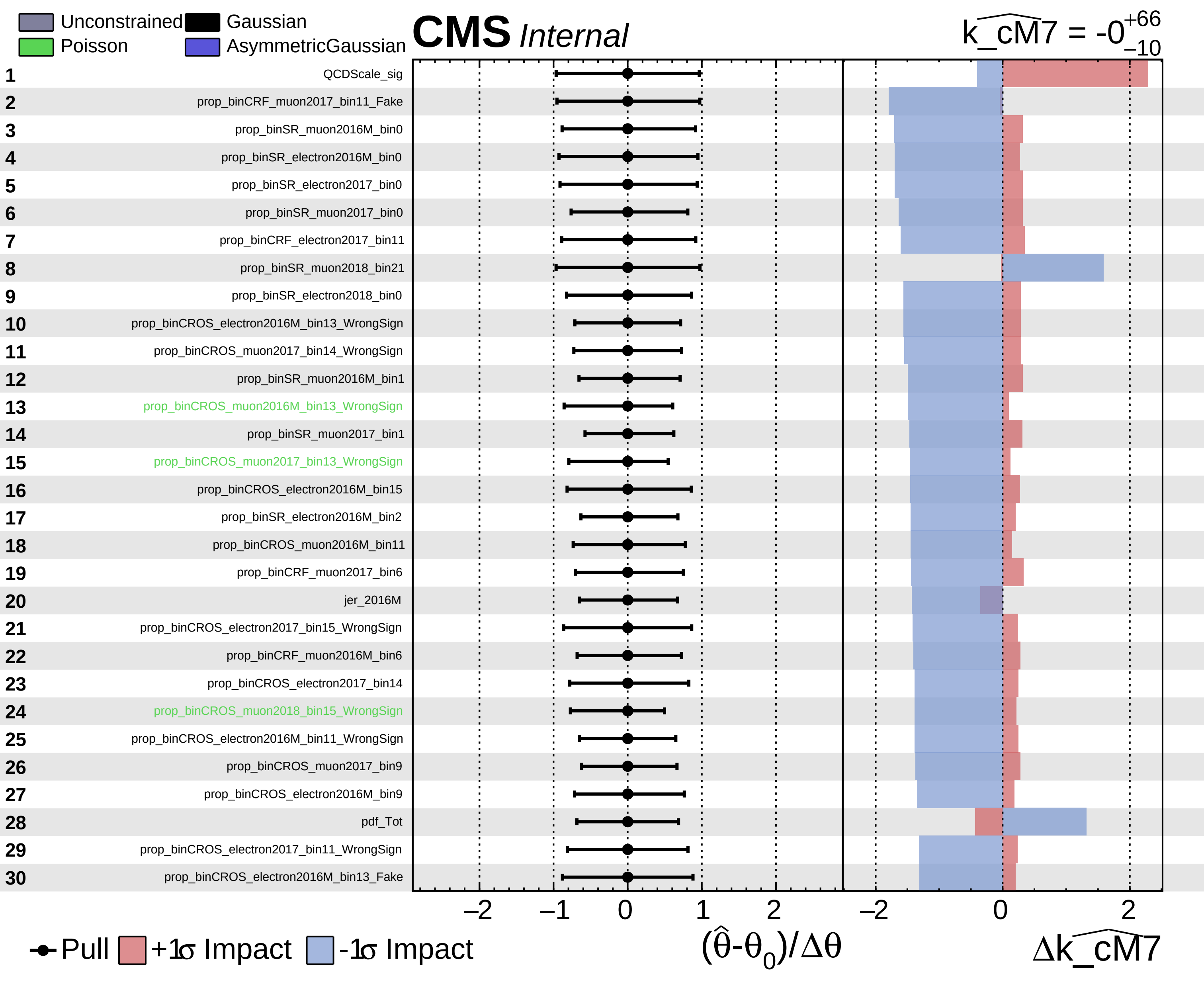
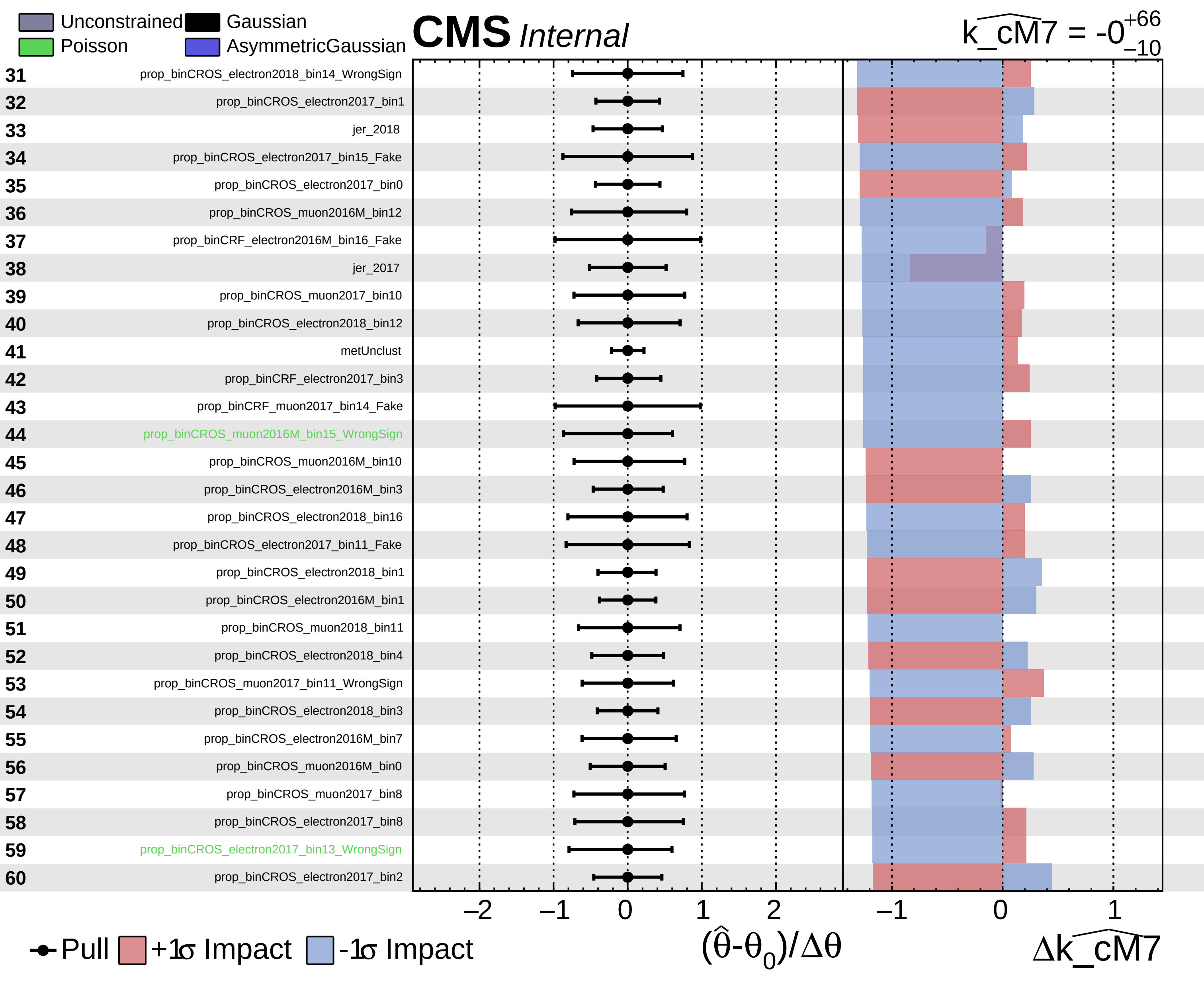
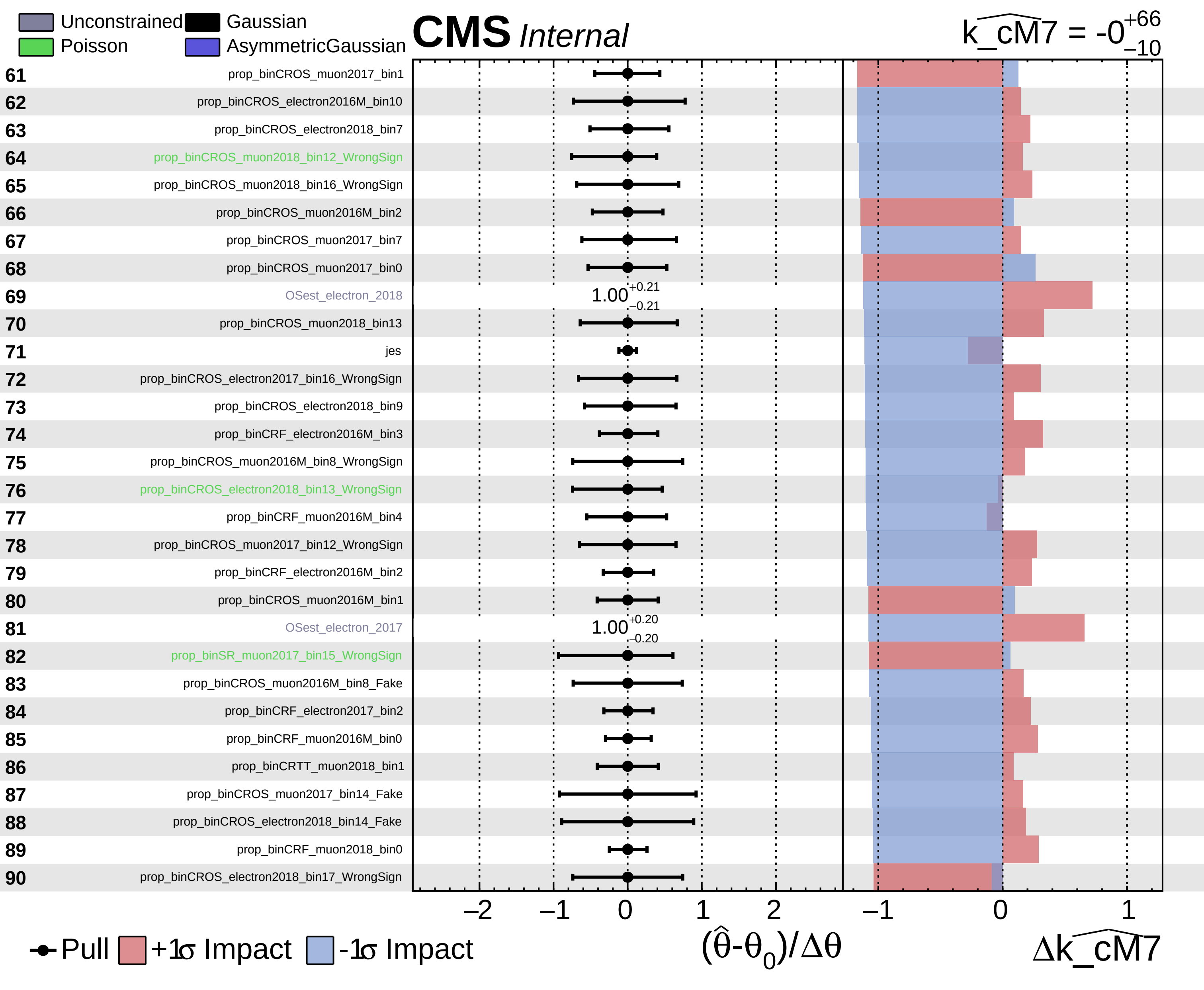
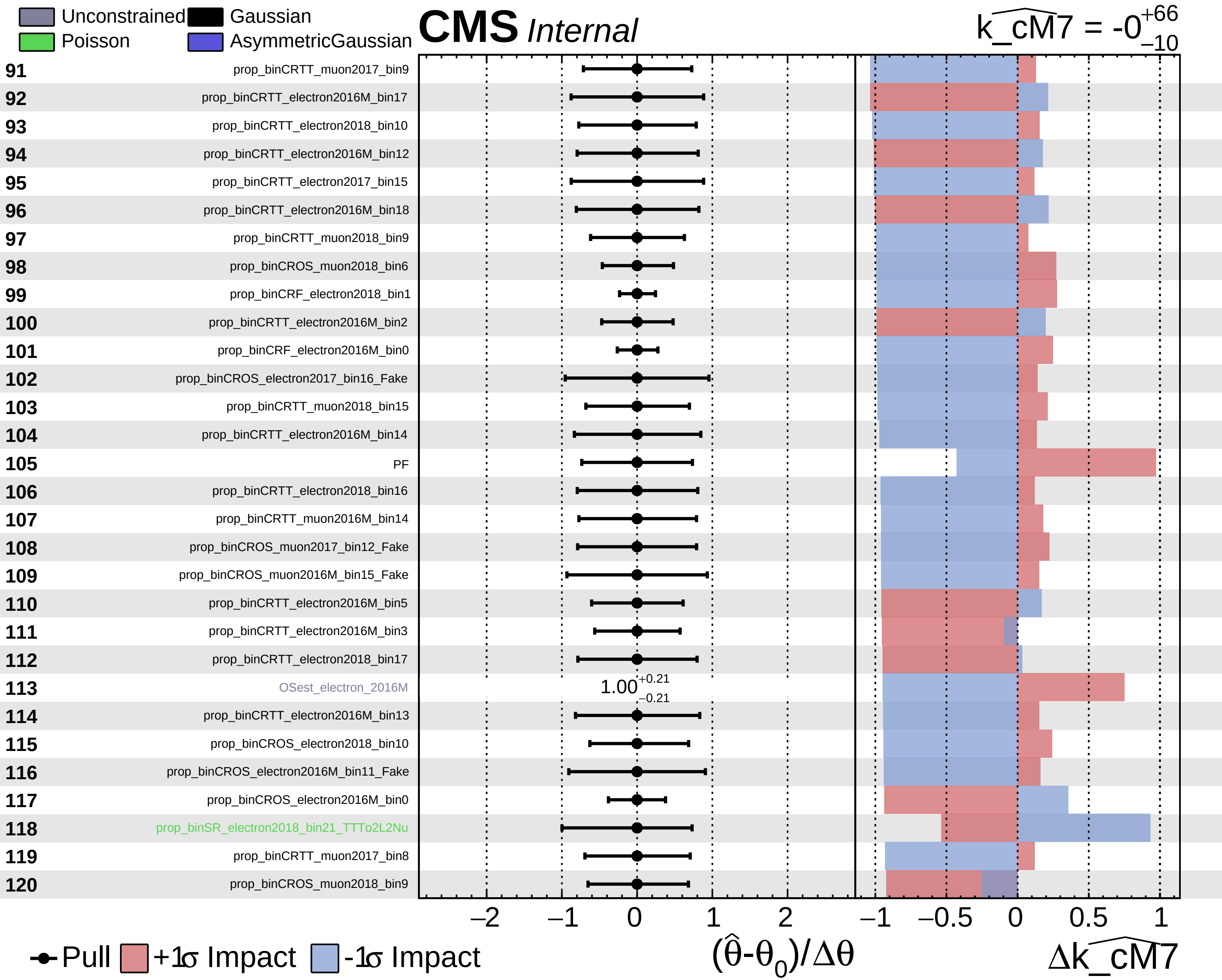


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM7 = -0 +66 −10
1 QCDScale_sig
2 prop_binCRF_muon2017_bin11_Fake
3 prop_binSR_muon2016M_bin0
4 prop_binSR_electron2016M_bin0
5 prop_binSR_electron2017_bin0
6 prop_binSR_muon2017_bin0
7 prop_binCRF_electron2017_bin11
8 prop_binSR_muon2018_bin21
9 prop_binSR_electron2018_bin0
10 prop_binCROS_electron2016M_bin13_WrongSign
11 prop_binCROS_muon2017_bin14_WrongSign
12 prop_binSR_muon2016M_bin1
13 prop_binCROS_muon2016M_bin13_WrongSign
14 prop_binSR_muon2017_bin1
15 prop_binCROS_muon2017_bin13_WrongSign
16 prop_binCROS_electron2016M_bin15
17 prop_binSR_electron2016M_bin2
18 prop_binCROS_muon2016M_bin11
19 prop_binCRF_muon2017_bin6
20 jer_2016M
21 prop_binCROS_electron2017_bin15_WrongSign
22 prop_binCRF_muon2016M_bin6
23 prop_binCROS_electron2017_bin14
24 prop_binCROS_muon2018_bin15_WrongSign
25 prop_binCROS_electron2016M_bin11_WrongSign
26 prop_binCROS_muon2017_bin9
27 prop_binCROS_electron2016M_bin9
28 pdf_Tot
29 prop_binCROS_electron2017_bin11_WrongSign
30 prop_binCROS_electron2016M_bin13_Fake
−2 −1 0 1 2
−2 0 2
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cM7

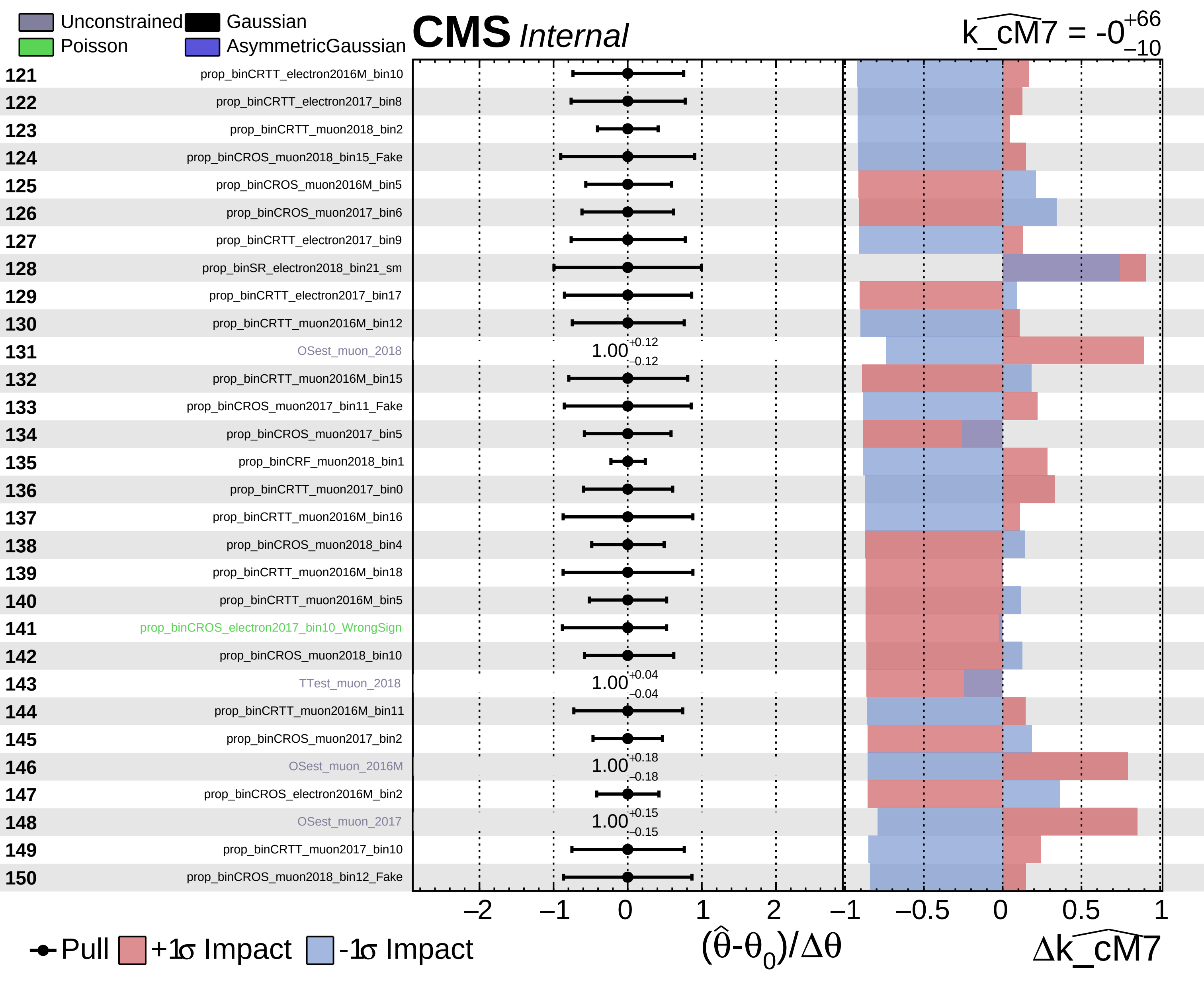

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM7 = -0$^{+66}_{-10}$
31 prop_binCROS_electron2018_bin14_WrongSign
32 prop_binCROS_electron2017_bin1
33 jer_2018
34 prop_binCROS_electron2017_bin15_Fake
35 prop_binCROS_electron2017_bin0
36 prop_binCROS_muon2016M_bin12
37 prop_binCRF_electron2016M_bin16_Fake
38 jer_2017
39 prop_binCROS_muon2017_bin10
40 prop_binCROS_electron2018_bin12
41 metUnclust
42 prop_binCRF_electron2017_bin3
43 prop_binCRF_muon2017_bin14_Fake
44 prop_binCROS_muon2016M_bin15_WrongSign
45 prop_binCROS_muon2016M_bin10
46 prop_binCROS_electron2016M_bin3
47 prop_binCROS_electron2018_bin16
48 prop_binCROS_electron2017_bin11_Fake
49 prop_binCROS_electron2018_bin1
50 prop_binCROS_electron2016M_bin1
51 prop_binCROS_muon2018_bin11
52 prop_binCROS_electron2018_bin4
53 prop_binCROS_muon2017_bin11_WrongSign
54 prop_binCROS_electron2018_bin3
55 prop_binCROS_electron2016M_bin7
56 prop_binCROS_muon2016M_bin0
57 prop_binCROS_muon2017_bin8
58 prop_binCROS_electron2017_bin8
59 prop_binCROS_electron2017_bin13_WrongSign
60 prop_binCROS_electron2017_bin2
−2
−1
0
1
2
−1
0
1
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cM7

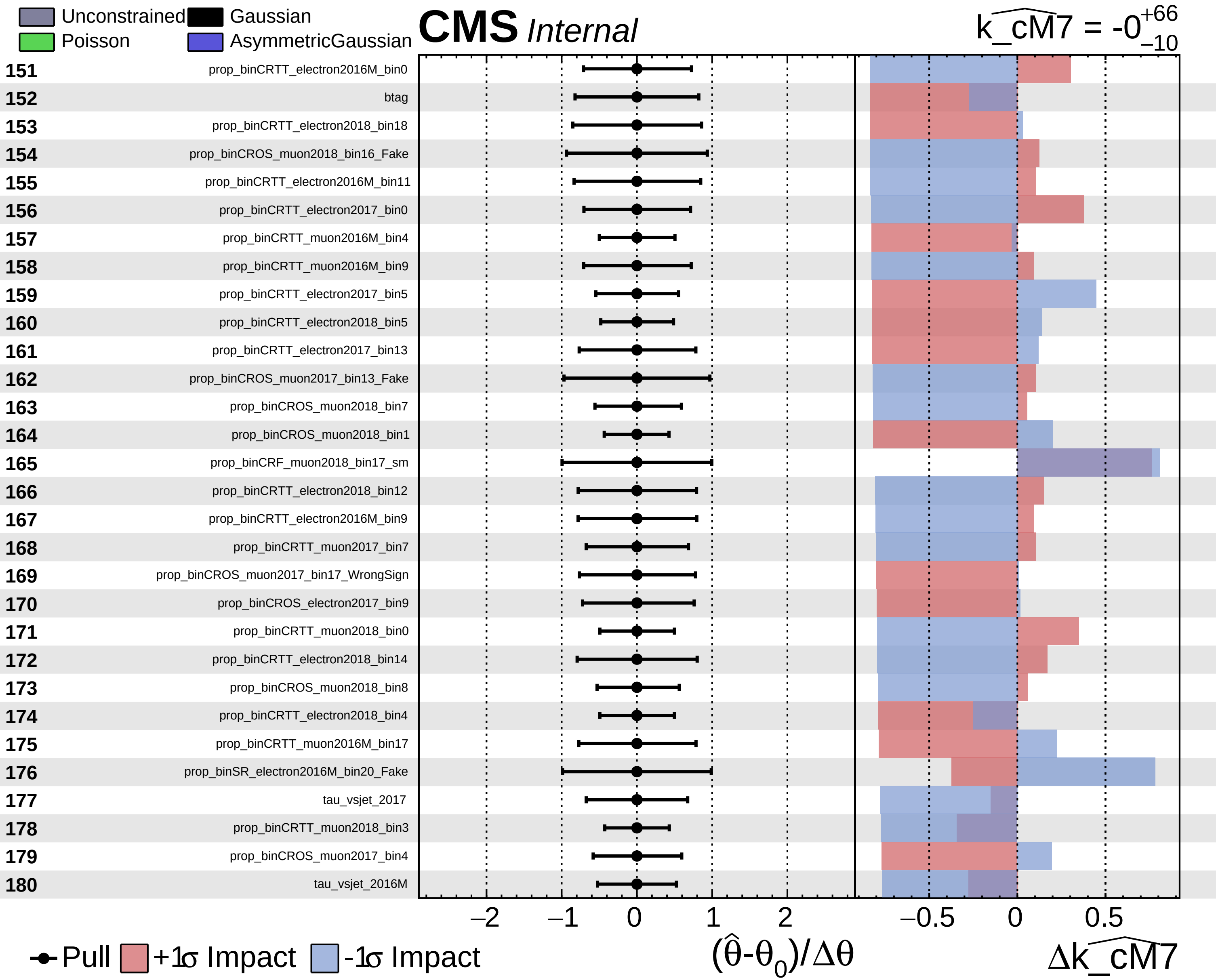


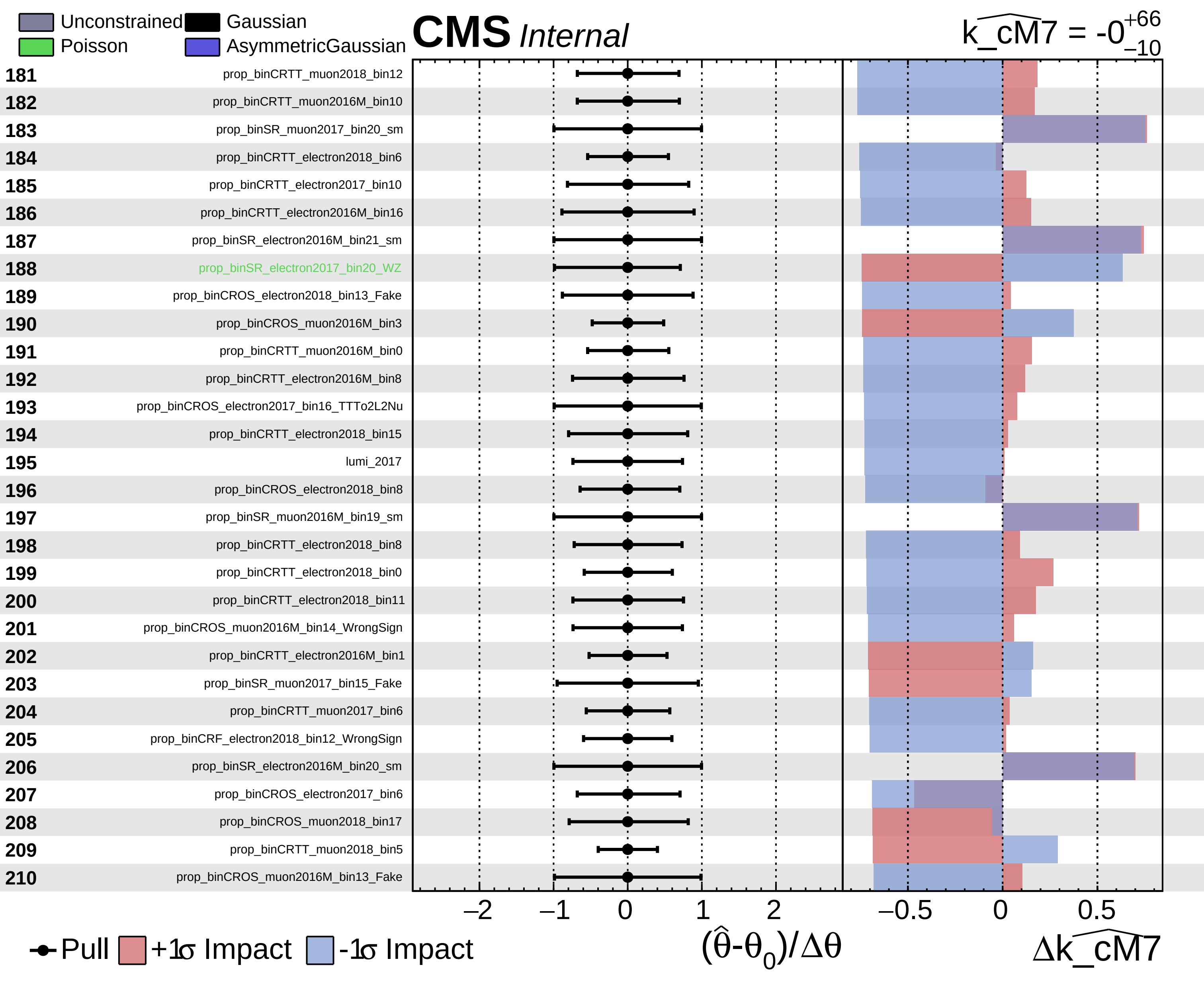
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM7 = -0 +66 −10
61 prop_binCROS_muon2017_bin1
62 prop_binCROS_electron2016M_bin10
63 prop_binCROS_electron2018_bin7
64 prop_binCROS_muon2018_bin12_WrongSign
65 prop_binCROS_muon2018_bin16_WrongSign
66 prop_binCROS_muon2016M_bin2
67 prop_binCROS_muon2017_bin7
68 prop_binCROS_muon2017_bin0
69 OSest_electron_2018
1.00 +0.21 −0.21
70 prop_binCROS_muon2018_bin13
71 jes
72 prop_binCROS_electron2017_bin16_WrongSign
73 prop_binCROS_electron2018_bin9
74 prop_binCRF_electron2016M_bin3
75 prop_binCROS_muon2016M_bin8_WrongSign
76 prop_binCROS_electron2018_bin13_WrongSign
77 prop_binCRF_muon2016M_bin4
78 prop_binCROS_muon2017_bin12_WrongSign
79 prop_binCRF_electron2016M_bin2
80 prop_binCROS_muon2016M_bin1
81 OSest_electron_2017
1.00 +0.20 −0.20
82 prop_binSR_muon2017_bin15_WrongSign
83 prop_binCROS_muon2016M_bin8_Fake
84 prop_binCRF_electron2017_bin2
85 prop_binCRF_muon2016M_bin0
86 prop_binCRTT_muon2018_bin1
87 prop_binCROS_muon2017_bin14_Fake
88 prop_binCROS_electron2018_bin14_Fake
89 prop_binCRF_muon2018_bin0
90 prop_binCROS_electron2018_bin17_WrongSign
−2 −1 0 1 2
−1 0 1
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cM7



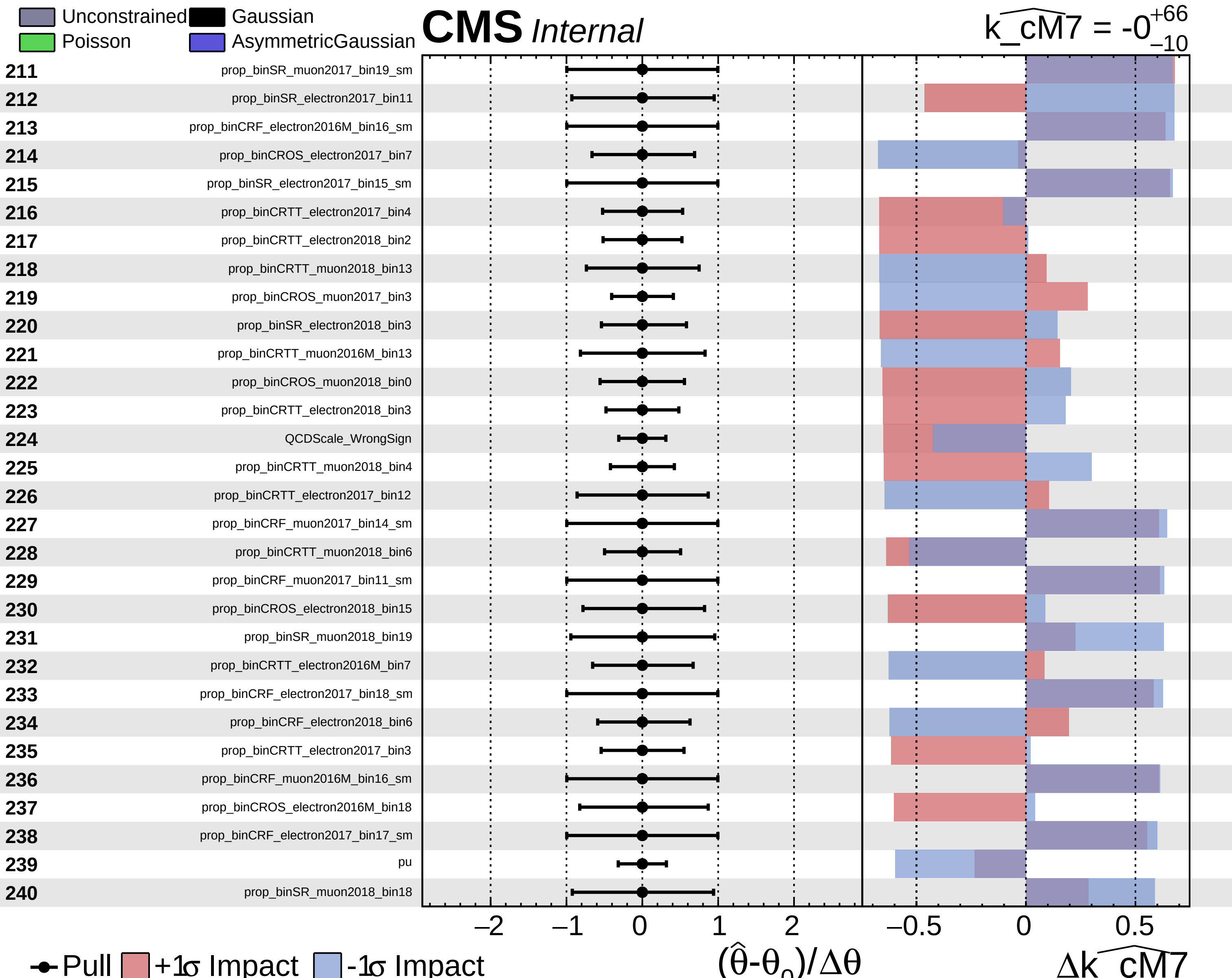
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM7 = -0 +66 −10
91 prop_binCRTT_muon2017_bin9
92 prop_binCRTT_electron2016M_bin17
93 prop_binCRTT_electron2018_bin10
94 prop_binCRTT_electron2016M_bin12
95 prop_binCRTT_electron2017_bin15
96 prop_binCRTT_electron2016M_bin18
97 prop_binCRTT_muon2018_bin9
98 prop_binCROS_muon2018_bin6
99 prop_binCRF_electron2018_bin1
100 prop_binCRTT_electron2016M_bin2
101 prop_binCRF_electron2016M_bin0
102 prop_binCROS_electron2017_bin16_Fake
103 prop_binCRTT_muon2018_bin15
104 prop_binCRTT_electron2016M_bin14
105 PF
106 prop_binCRTT_electron2018_bin16
107 prop_binCRTT_muon2016M_bin14
108 prop_binCROS_muon2017_bin12_Fake
109 prop_binCROS_muon2016M_bin15_Fake
110 prop_binCRTT_electron2016M_bin5
111 prop_binCRTT_electron2016M_bin3
112 prop_binCRTT_electron2018_bin17
113 OSest_electron_2016M
1.00 +0.21 −0.21
114 prop_binCRTT_electron2016M_bin13
115 prop_binCROS_electron2018_bin10
116 prop_binCROS_electron2016M_bin11_Fake
117 prop_binCROS_electron2016M_bin0
118 prop_binSR_electron2018_bin21_TTTo2L2Nu
119 prop_binCRTT_muon2017_bin8
120 prop_binCROS_muon2018_bin9
−2 −1 0 1 2
−1 −0.5 0 0.5 1
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cM7


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM7 = -0 +66 −10
121 prop_binCRTT_electron2016M_bin10
122 prop_binCRTT_electron2017_bin8
123 prop_binCRTT_muon2018_bin2
124 prop_binCROS_muon2018_bin15_Fake
125 prop_binCROS_muon2016M_bin5
126 prop_binCROS_muon2017_bin6
127 prop_binCRTT_electron2017_bin9
128 prop_binSR_electron2018_bin21_sm
129 prop_binCRTT_electron2017_bin17
130 prop_binCRTT_muon2016M_bin12
131 OSest_muon_2018
1.00 +0.12 −0.12
132 prop_binCRTT_muon2016M_bin15
133 prop_binCROS_muon2017_bin11_Fake
134 prop_binCROS_muon2017_bin5
135 prop_binCRF_muon2018_bin1
136 prop_binCRTT_muon2017_bin0
137 prop_binCRTT_muon2016M_bin16
138 prop_binCROS_muon2018_bin4
139 prop_binCRTT_muon2016M_bin18
140 prop_binCRTT_muon2016M_bin5
141 prop_binCROS_electron2017_bin10_WrongSign
142 prop_binCROS_muon2018_bin10
143 TTest_muon_2018
1.00 +0.04 −0.04
144 prop_binCRTT_muon2016M_bin11
145 prop_binCROS_muon2017_bin2
146 OSest_muon_2016M
1.00 +0.18 −0.18
147 prop_binCROS_electron2016M_bin2
148 OSest_muon_2017
1.00 +0.15 −0.15
149 prop_binCRTT_muon2017_bin10
150 prop_binCROS_muon2018_bin12_Fake
−2 −1 0 1 2
−1 −0.5 0 0.5 1
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cM7


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM7 = -0+66 −10
151 prop_binCRTT_electron2016M_bin0
152 btag
153 prop_binCRTT_electron2018_bin18
154 prop_binCROS_muon2018_bin16_Fake
155 prop_binCRTT_electron2016M_bin11
156 prop_binCRTT_electron2017_bin0
157 prop_binCRTT_muon2016M_bin4
158 prop_binCRTT_muon2016M_bin9
159 prop_binCRTT_electron2017_bin5
160 prop_binCRTT_electron2018_bin5
161 prop_binCRTT_electron2017_bin13
162 prop_binCROS_muon2017_bin13_Fake
163 prop_binCROS_muon2018_bin7
164 prop_binCROS_muon2018_bin1
165 prop_binCRF_muon2018_bin17_sm
166 prop_binCRTT_electron2018_bin12
167 prop_binCRTT_electron2016M_bin9
168 prop_binCRTT_muon2017_bin7
169 prop_binCROS_muon2017_bin17_WrongSign
170 prop_binCROS_electron2017_bin9
171 prop_binCRTT_muon2018_bin0
172 prop_binCRTT_electron2018_bin14
173 prop_binCROS_muon2018_bin8
174 prop_binCRTT_electron2018_bin4
175 prop_binCRTT_muon2016M_bin17
176 prop_binSR_electron2016M_bin20_Fake
177 tau_vsjet_2017
178 prop_binCRTT_muon2018_bin3
179 prop_binCROS_muon2017_bin4
180 tau_vsjet_2016M
−2
−1
0
1
2
−0.5
0
0.5
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cM7



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM7} = -0^{+66}_{-10}$
181 prop_binCRTT_muon2018_bin12
182 prop_binCRTT_muon2016M_bin10
183 prop_binSR_muon2017_bin20_sm
184 prop_binCRTT_electron2018_bin6
185 prop_binCRTT_electron2017_bin10
186 prop_binCRTT_electron2016M_bin16
187 prop_binSR_electron2016M_bin21_sm
188 prop_binSR_electron2017_bin20_WZ
189 prop_binCROS_electron2018_bin13_Fake
190 prop_binCROS_muon2016M_bin3
191 prop_binCRTT_muon2016M_bin0
192 prop_binCRTT_electron2016M_bin8
193 prop_binCROS_electron2017_bin16_TTTo2L2Nu
194 prop_binCRTT_electron2018_bin15
195 lumi_2017
196 prop_binCROS_electron2018_bin8
197 prop_binSR_muon2016M_bin19_sm
198 prop_binCRTT_electron2018_bin8
199 prop_binCRTT_electron2018_bin0
200 prop_binCRTT_electron2018_bin11
201 prop_binCROS_muon2016M_bin14_WrongSign
202 prop_binCRTT_electron2016M_bin1
203 prop_binSR_muon2017_bin15_Fake
204 prop_binCRTT_muon2017_bin6
205 prop_binCRF_electron2018_bin12_WrongSign
206 prop_binSR_electron2016M_bin20_sm
207 prop_binCROS_electron2017_bin6
208 prop_binCROS_muon2018_bin17
209 prop_binCRTT_muon2018_bin5
210 prop_binCROS_muon2016M_bin13_Fake
−2 −1 0 1 2
−0.5 0 0.5
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM7}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM7} = -0^{+66}_{-10}$
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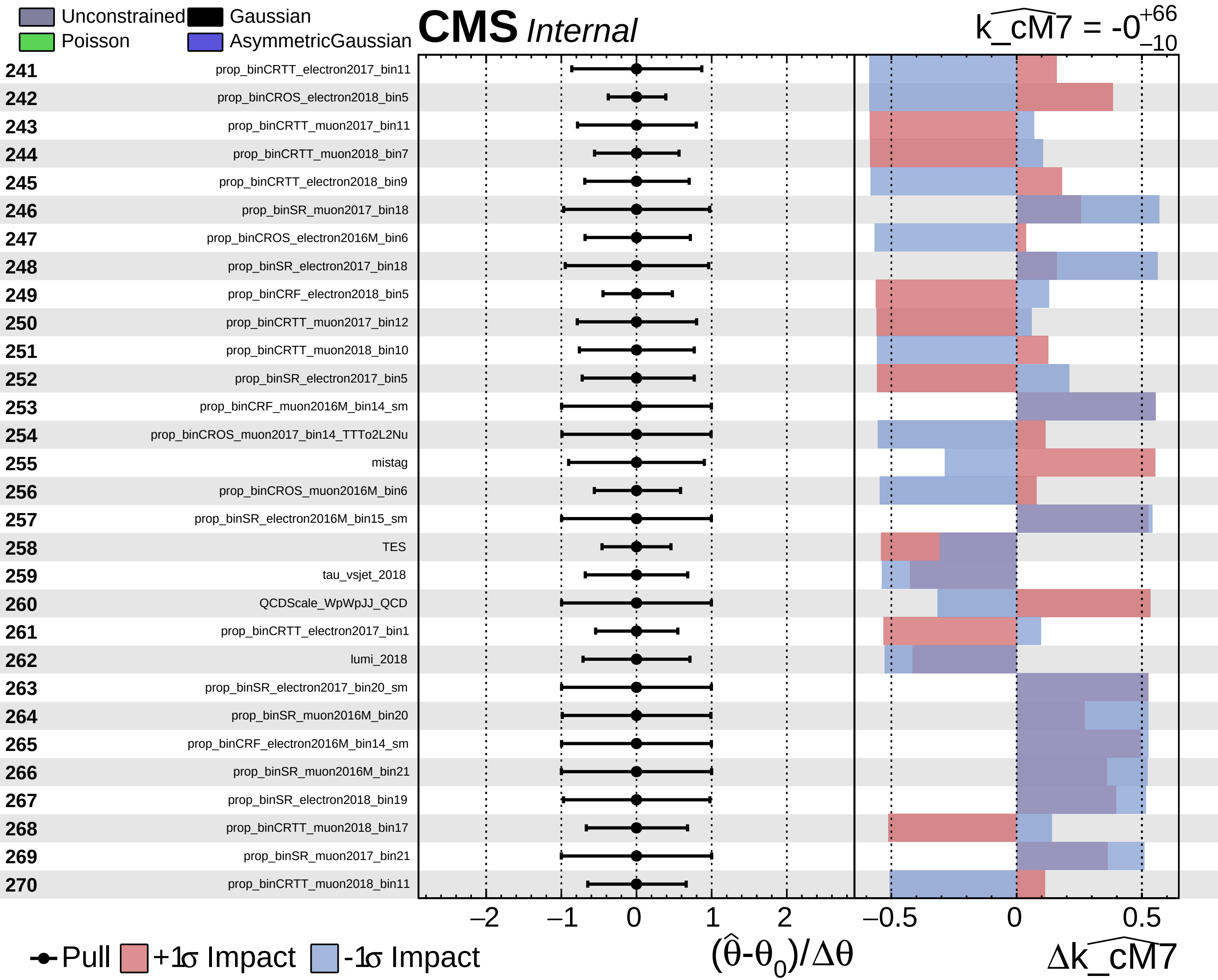
212 prop_binSR_electron2017_bin11
213 prop_binCRF_electron2016M_bin16_sm
214 prop_binCROS_electron2017_bin7
215 prop_binSR_electron2017_bin15_sm
216 prop_binCRTT_electron2017_bin4
217 prop_binCRTT_electron2018_bin2
218 prop_binCRTT_muon2018_bin13
219 prop_binCROS_muon2017_bin3
220 prop_binSR_electron2018_bin3
221 prop_binCRTT_muon2016M_bin13
222 prop_binCROS_muon2018_bin0
223 prop_binCRTT_electron2018_bin3
224 QCDScale_WrongSign
225 prop_binCRTT_muon2018_bin4
226 prop_binCRTT_electron2017_bin12
227 prop_binCRF_muon2017_bin14_sm
228 prop_binCRTT_muon2018_bin6
229 prop_binCRF_muon2017_bin11_sm
230 prop_binCROS_electron2018_bin15
231 prop_binSR_muon2018_bin19
232 prop_binCRTT_electron2016M_bin7
233 prop_binCRF_electron2017_bin18_sm
234 prop_binCRF_electron2018_bin6
235 prop_binCRTT_electron2017_bin3
236 prop_binCRF_muon2016M_bin16_sm
237 prop_binCROS_electron2016M_bin18
238 prop_binCRF_electron2017_bin17_sm
239 pu
240 prop_binSR_muon2018_bin18
−2 −1 0 1 2
−0.5 0 0.5
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM7}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM7 = -0 +66 −10
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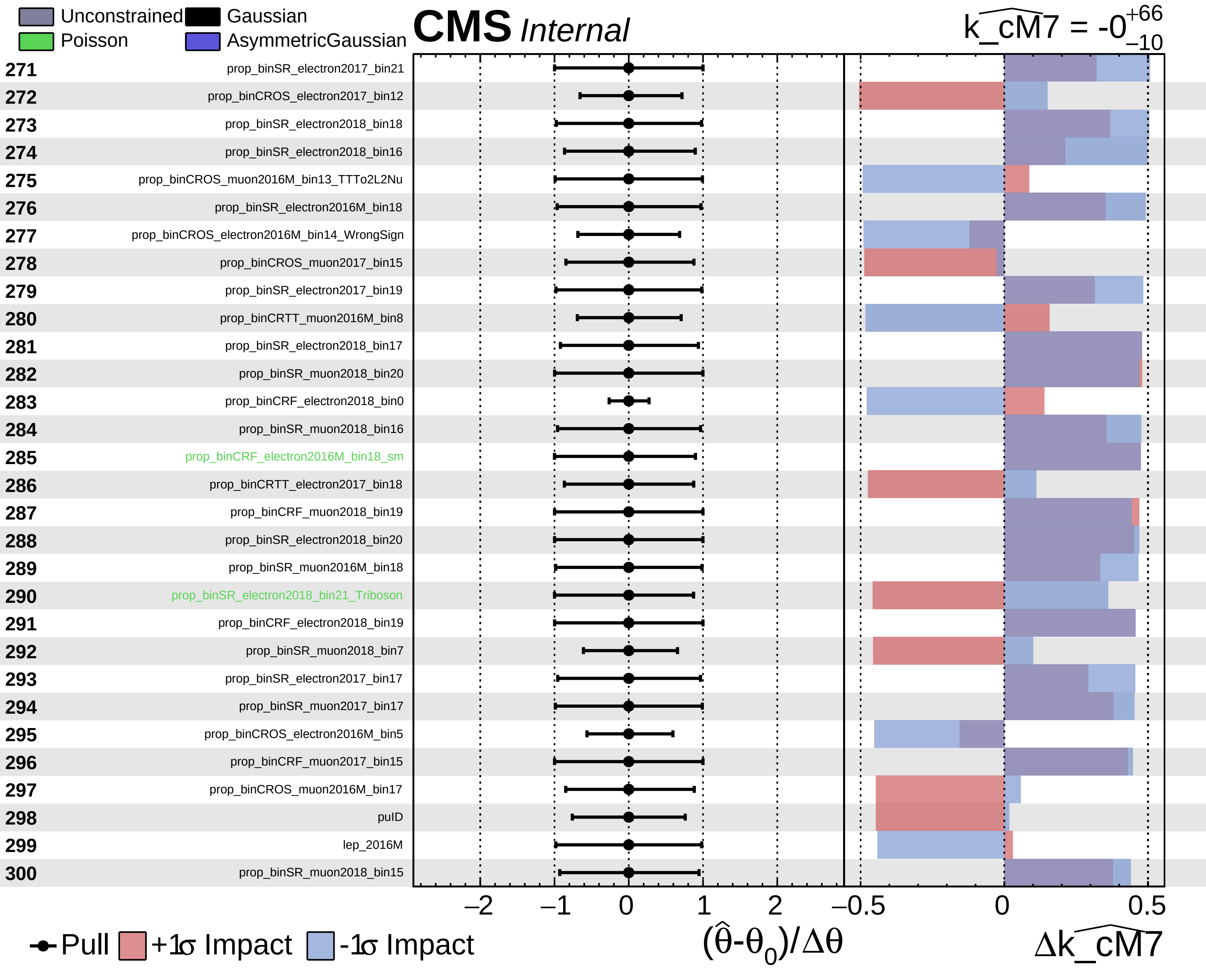
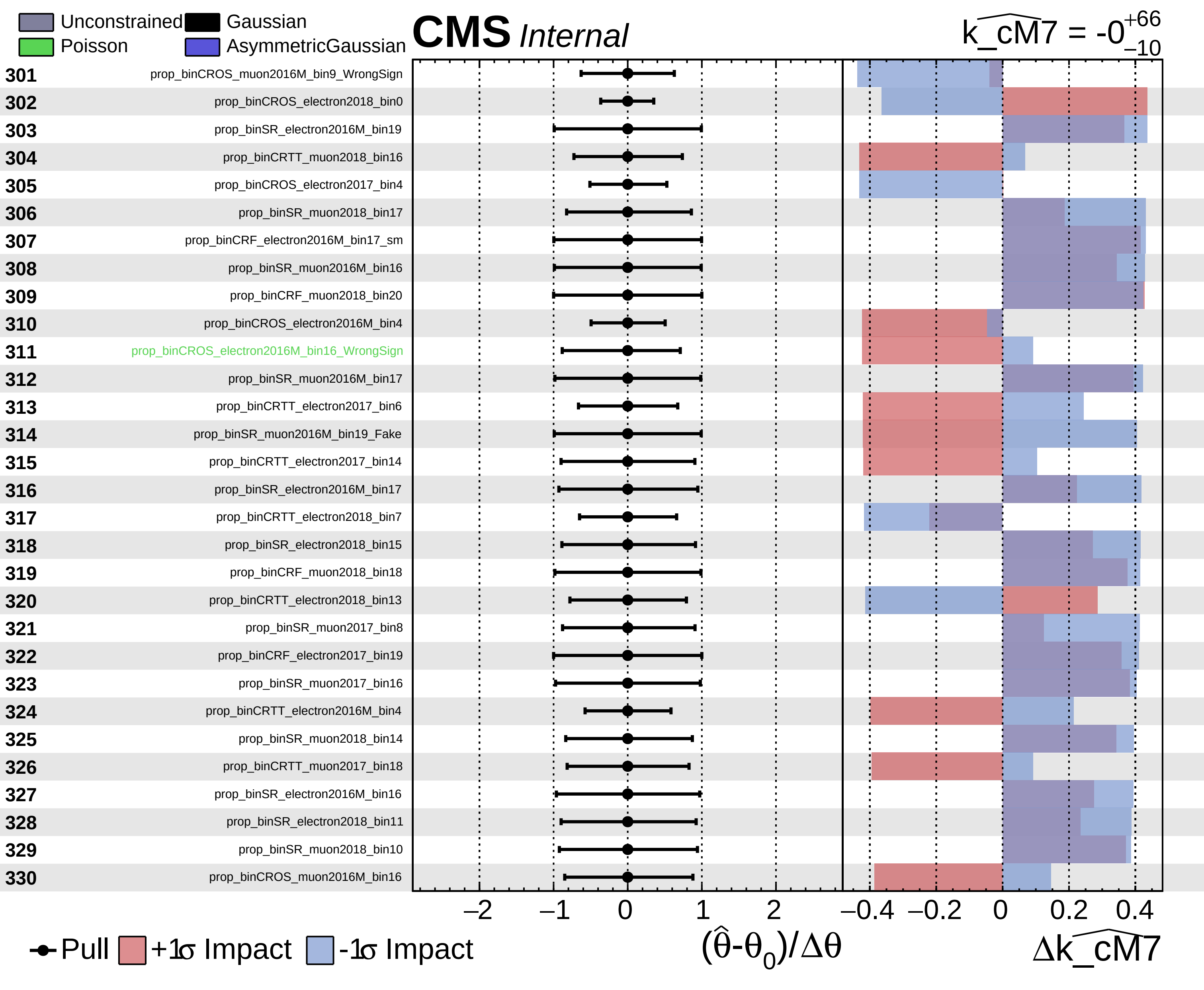
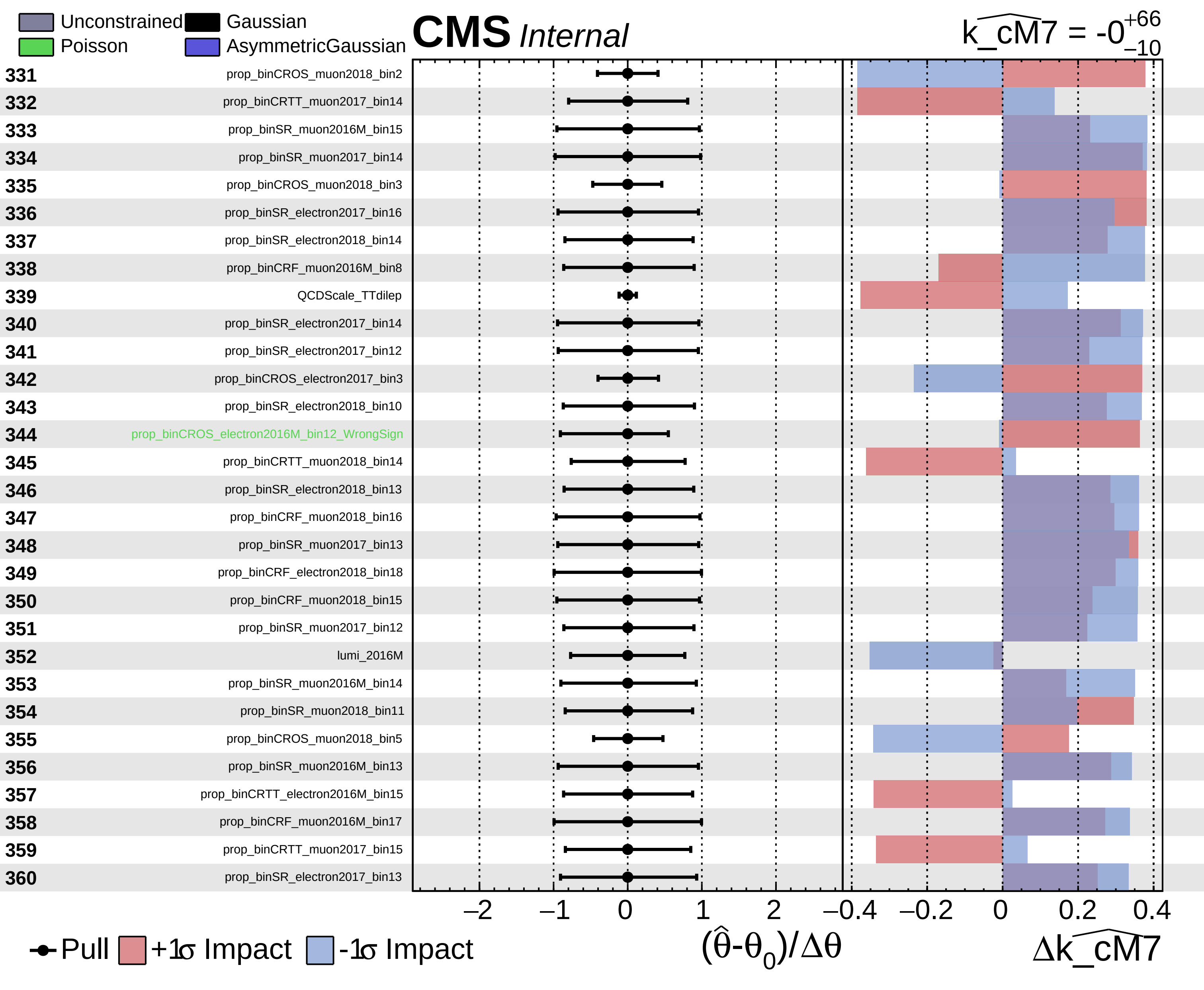
242 prop_binCROS_electron2018_bin5
243 prop_binCRTT_muon2017_bin11
244 prop_binCRTT_muon2018_bin7
245 prop_binCRTT_electron2018_bin9
246 prop_binSR_muon2017_bin18
247 prop_binCROS_electron2016M_bin6
248 prop_binSR_electron2017_bin18
249 prop_binCRF_electron2018_bin5
250 prop_binCRTT_muon2017_bin12
251 prop_binCRTT_muon2018_bin10
252 prop_binSR_electron2017_bin5
253 prop_binCRF_muon2016M_bin14_sm
254 prop_binCROS_muon2017_bin14_TTTo2L2Nu
255 mistag
256 prop_binCROS_muon2016M_bin6
257 prop_binSR_electron2016M_bin15_sm
258 TES
259 tau_vsjet_2018
260 QCDScale_WpWpJJ_QCD
261 prop_binCRTT_electron2017_bin1
262 lumi_2018
263 prop_binSR_electron2017_bin20_sm
264 prop_binSR_muon2016M_bin20
265 prop_binCRF_electron2016M_bin14_sm
266 prop_binSR_muon2016M_bin21
267 prop_binSR_electron2018_bin19
268 prop_binCRTT_muon2018_bin17
269 prop_binSR_muon2017_bin21
270 prop_binCRTT_muon2018_bin11
−2
−1
0
1
2
−0.5
0
0.5
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cM7



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM7 = -0$^{+66}_{-10}$
271 prop_binSR_electron2017_bin21
272 prop_binCROS_electron2017_bin12
273 prop_binSR_electron2018_bin18
274 prop_binSR_electron2018_bin16
275 prop_binCROS_muon2016M_bin13_TTTo2L2Nu
276 prop_binSR_electron2016M_bin18
277 prop_binCROS_electron2016M_bin14_WrongSign
278 prop_binCROS_muon2017_bin15
279 prop_binSR_electron2017_bin19
280 prop_binCRTT_muon2016M_bin8
281 prop_binSR_electron2018_bin17
282 prop_binSR_muon2018_bin20
283 prop_binCRF_electron2018_bin0
284 prop_binSR_muon2018_bin16
285 prop_binCRF_electron2016M_bin18_sm
286 prop_binCRTT_electron2017_bin18
287 prop_binCRF_muon2018_bin19
288 prop_binSR_electron2018_bin20
289 prop_binSR_muon2016M_bin18
290 prop_binSR_electron2018_bin21_Triboson
291 prop_binCRF_electron2018_bin19
292 prop_binSR_muon2018_bin7
293 prop_binSR_electron2017_bin17
294 prop_binSR_muon2017_bin17
295 prop_binCROS_electron2016M_bin5
296 prop_binCRF_muon2017_bin15
297 prop_binCROS_muon2016M_bin17
298 puID
299 lep_2016M
300 prop_binSR_muon2018_bin15
−2 −1 0 1 2
−0.5 0 0.5
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cM7



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM7} = -0^{+66}_{-10}$
301 prop_binCROS_muon2016M_bin9_WrongSign
302 prop_binCROS_electron2018_bin0
303 prop_binSR_electron2016M_bin19
304 prop_binCRTT_muon2018_bin16
305 prop_binCROS_electron2017_bin4
306 prop_binSR_muon2018_bin17
307 prop_binCRF_electron2016M_bin17_sm
308 prop_binSR_muon2016M_bin16
309 prop_binCRF_muon2018_bin20
310 prop_binCROS_electron2016M_bin4
311 prop_binCROS_electron2016M_bin16_WrongSign
312 prop_binSR_muon2016M_bin17
313 prop_binCRTT_electron2017_bin6
314 prop_binSR_muon2016M_bin19_Fake
315 prop_binCRTT_electron2017_bin14
316 prop_binSR_electron2016M_bin17
317 prop_binCRTT_electron2018_bin7
318 prop_binSR_electron2018_bin15
319 prop_binCRF_muon2018_bin18
320 prop_binCRTT_electron2018_bin13
321 prop_binSR_muon2017_bin8
322 prop_binCRF_electron2017_bin19
323 prop_binSR_muon2017_bin16
324 prop_binCRTT_electron2016M_bin4
325 prop_binSR_muon2018_bin14
326 prop_binCRTT_muon2017_bin18
327 prop_binSR_electron2016M_bin16
328 prop_binSR_electron2018_bin11
329 prop_binSR_muon2018_bin10
330 prop_binCROS_muon2016M_bin16
−2 −1 0 1 2
−0.4 −0.2 0 0.2 0.4
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM7}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM7} = -0^{+66}_{-10}$
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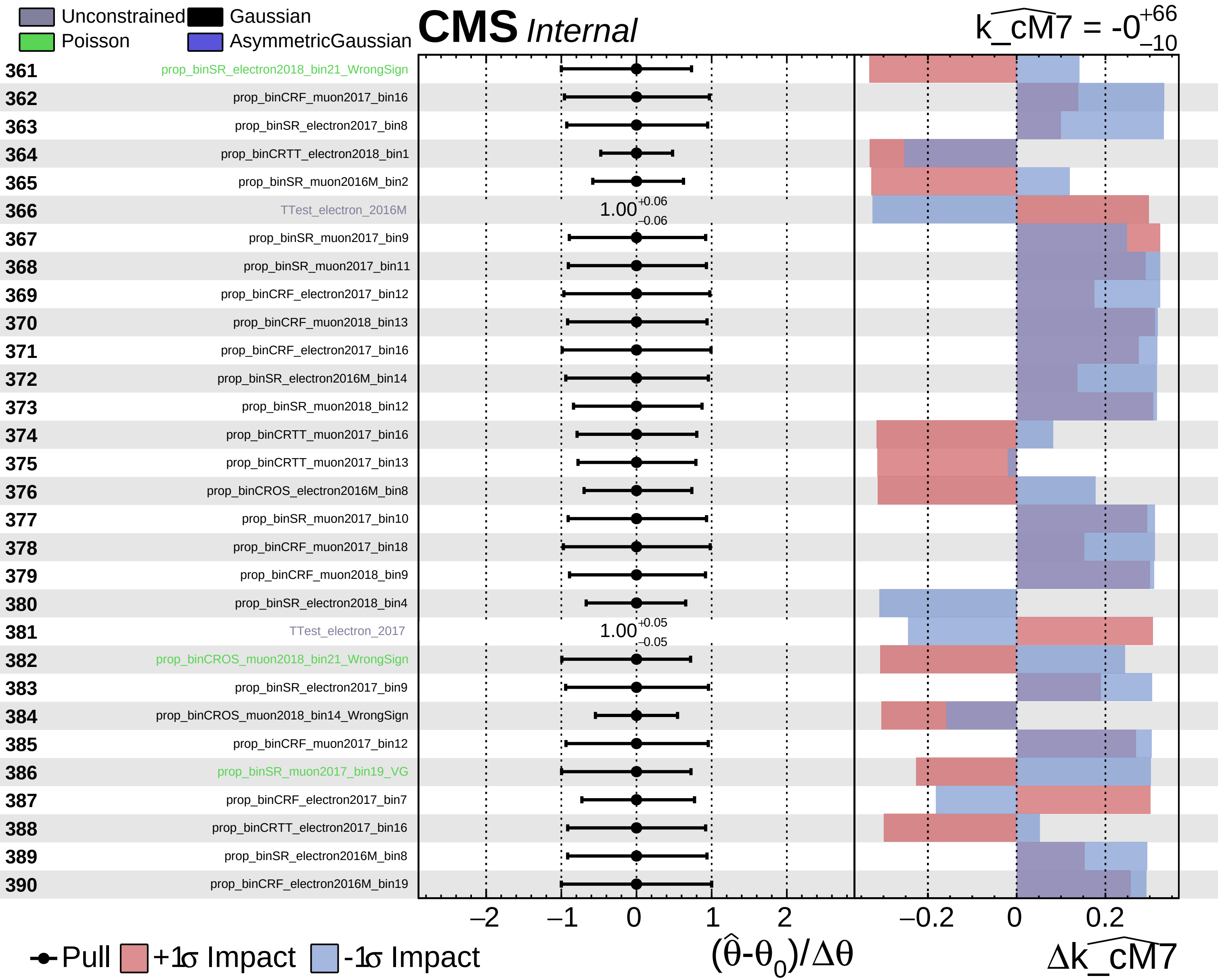
332 prop_binCRTT_muon2017_bin14
333 prop_binSR_muon2016M_bin15
334 prop_binSR_muon2017_bin14
335 prop_binCROS_muon2018_bin3
336 prop_binSR_electron2017_bin16
337 prop_binSR_electron2018_bin14
338 prop_binCRF_muon2016M_bin8
339 QCDScale_TTdilep
340 prop_binSR_electron2017_bin14
341 prop_binSR_electron2017_bin12
342 prop_binCROS_electron2017_bin3
343 prop_binSR_electron2018_bin10
344 prop_binCROS_electron2016M_bin12_WrongSign
345 prop_binCRTT_muon2018_bin14
346 prop_binSR_electron2018_bin13
347 prop_binCRF_muon2018_bin16
348 prop_binSR_muon2017_bin13
349 prop_binCRF_electron2018_bin18
350 prop_binCRF_muon2018_bin15
351 prop_binSR_muon2017_bin12
352 lumi_2016M
353 prop_binSR_muon2016M_bin14
354 prop_binSR_muon2018_bin11
355 prop_binCROS_muon2018_bin5
356 prop_binSR_muon2016M_bin13
357 prop_binCRTT_electron2016M_bin15
358 prop_binCRF_muon2016M_bin17
359 prop_binCRTT_muon2017_bin15
360 prop_binSR_electron2017_bin13
−2 −1 0 1 2
−0.4 −0.2 0 0.2 0.4
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM7}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM7} = -0^{+66}_{-10}$
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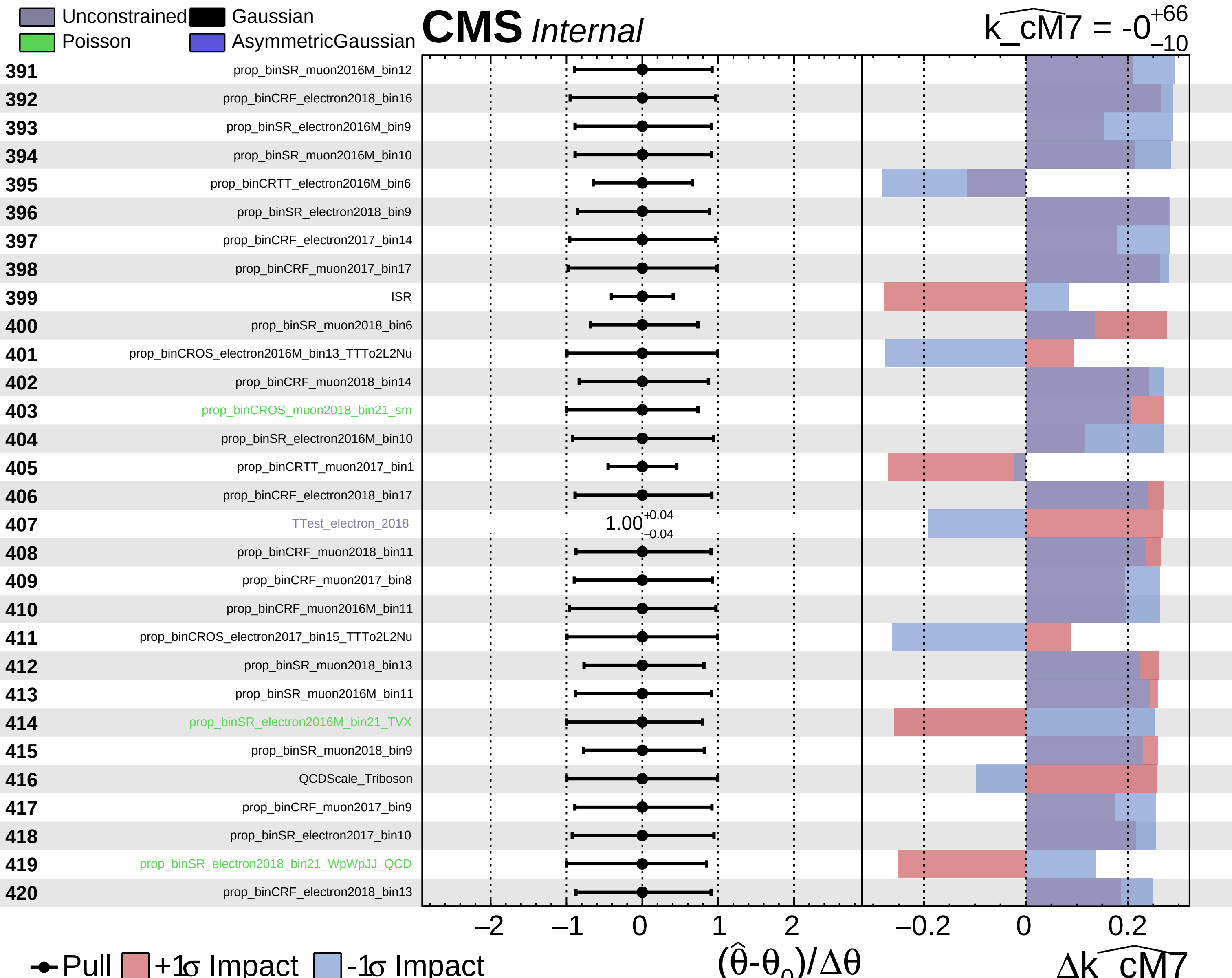
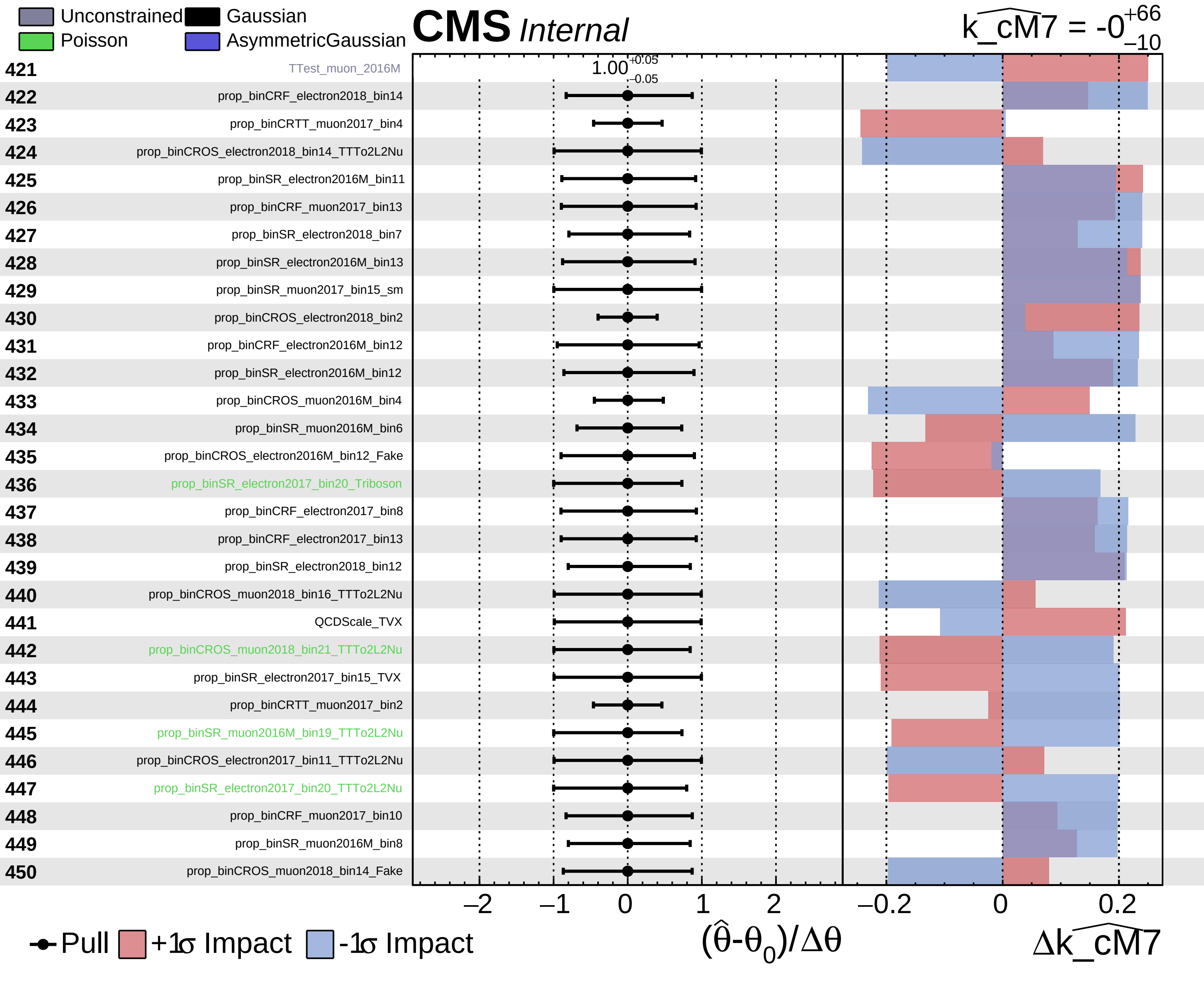
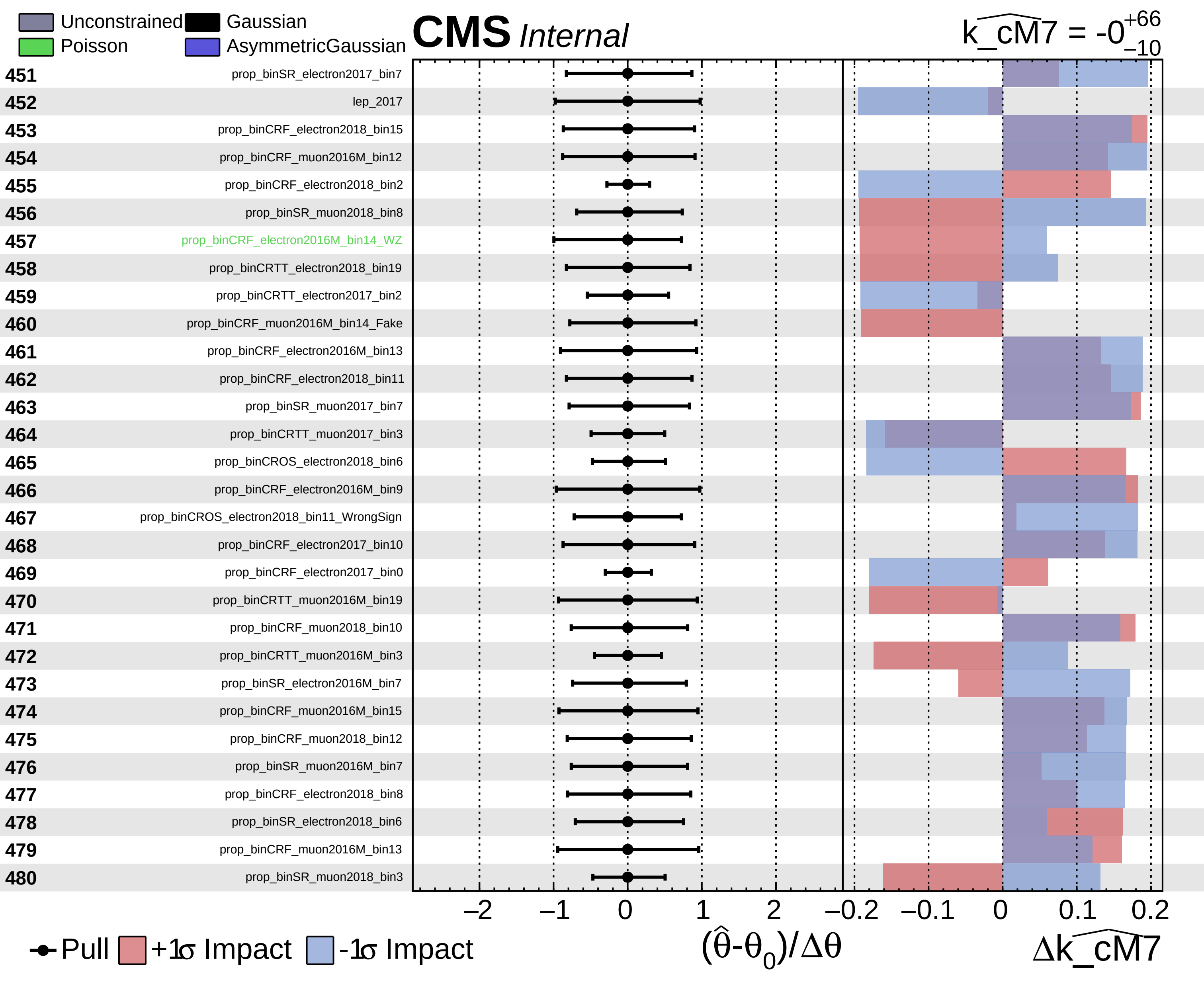
362 prop_binCRF_muon2017_bin16
363 prop_binSR_electron2017_bin8
364 prop_binCRTT_electron2018_bin1
365 prop_binSR_muon2016M_bin2
366 TTest_electron_2016M
$1.00^{+0.06}_{-0.06}$
367 prop_binSR_muon2017_bin9
368 prop_binSR_muon2017_bin11
369 prop_binCRF_electron2017_bin12
370 prop_binCRF_muon2018_bin13
371 prop_binCRF_electron2017_bin16
372 prop_binSR_electron2016M_bin14
373 prop_binSR_muon2018_bin12
374 prop_binCRTT_muon2017_bin16
375 prop_binCRTT_muon2017_bin13
376 prop_binCROS_electron2016M_bin8
377 prop_binSR_muon2017_bin10
378 prop_binCRF_muon2017_bin18
379 prop_binCRF_muon2018_bin9
380 prop_binSR_electron2018_bin4
381 TTest_electron_2017
$1.00^{+0.05}_{-0.05}$
382 prop_binCROS_muon2018_bin21_WrongSign
383 prop_binSR_electron2017_bin9
384 prop_binCROS_muon2018_bin14_WrongSign
385 prop_binCRF_muon2017_bin12
386 prop_binSR_muon2017_bin19_VG
387 prop_binCRF_electron2017_bin7
388 prop_binCRTT_electron2017_bin16
389 prop_binSR_electron2016M_bin8
390 prop_binCRF_electron2016M_bin19
−2 −1 0 1 2
−0.2 0 0.2
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM7}$



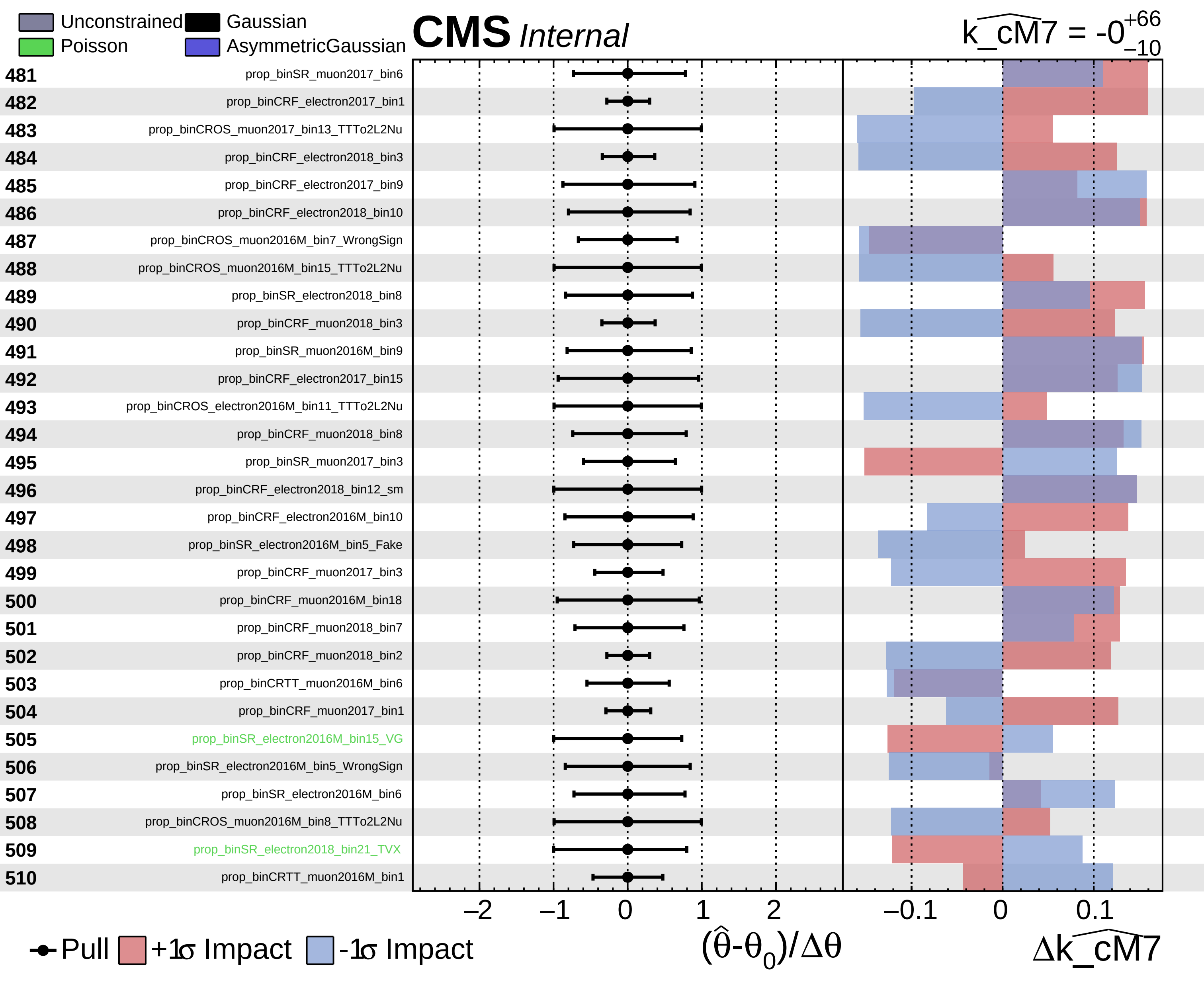
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM7} = -0^{+66}_{-10}$
391 prop_binSR_muon2016M_bin12
392 prop_binCRF_electron2018_bin16
393 prop_binSR_electron2016M_bin9
394 prop_binSR_muon2016M_bin10
395 prop_binCRTT_electron2016M_bin6
396 prop_binSR_electron2018_bin9
397 prop_binCRF_electron2017_bin14
398 prop_binCRF_muon2017_bin17
399 ISR
400 prop_binSR_muon2018_bin6
401 prop_binCROS_electron2016M_bin13_TTTo2L2Nu
402 prop_binCRF_muon2018_bin14
403 prop_binCROS_muon2018_bin21_sm
404 prop_binSR_electron2016M_bin10
405 prop_binCRTT_muon2017_bin1
406 prop_binCRF_electron2018_bin17
407 TTest_electron_2018
$1.00^{+0.04}_{-0.04}$
408 prop_binCRF_muon2018_bin11
409 prop_binCRF_muon2017_bin8
410 prop_binCRF_muon2016M_bin11
411 prop_binCROS_electron2017_bin15_TTTo2L2Nu
412 prop_binSR_muon2018_bin13
413 prop_binSR_muon2016M_bin11
414 prop_binSR_electron2016M_bin21_TVX
415 prop_binSR_muon2018_bin9
416 QCDScale_Triboson
417 prop_binCRF_muon2017_bin9
418 prop_binSR_electron2017_bin10
419 prop_binSR_electron2018_bin21_WpWpJJ_QCD
420 prop_binCRF_electron2018_bin13
−2
−1
0
1
2
−0.2
0
0.2
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM7}$



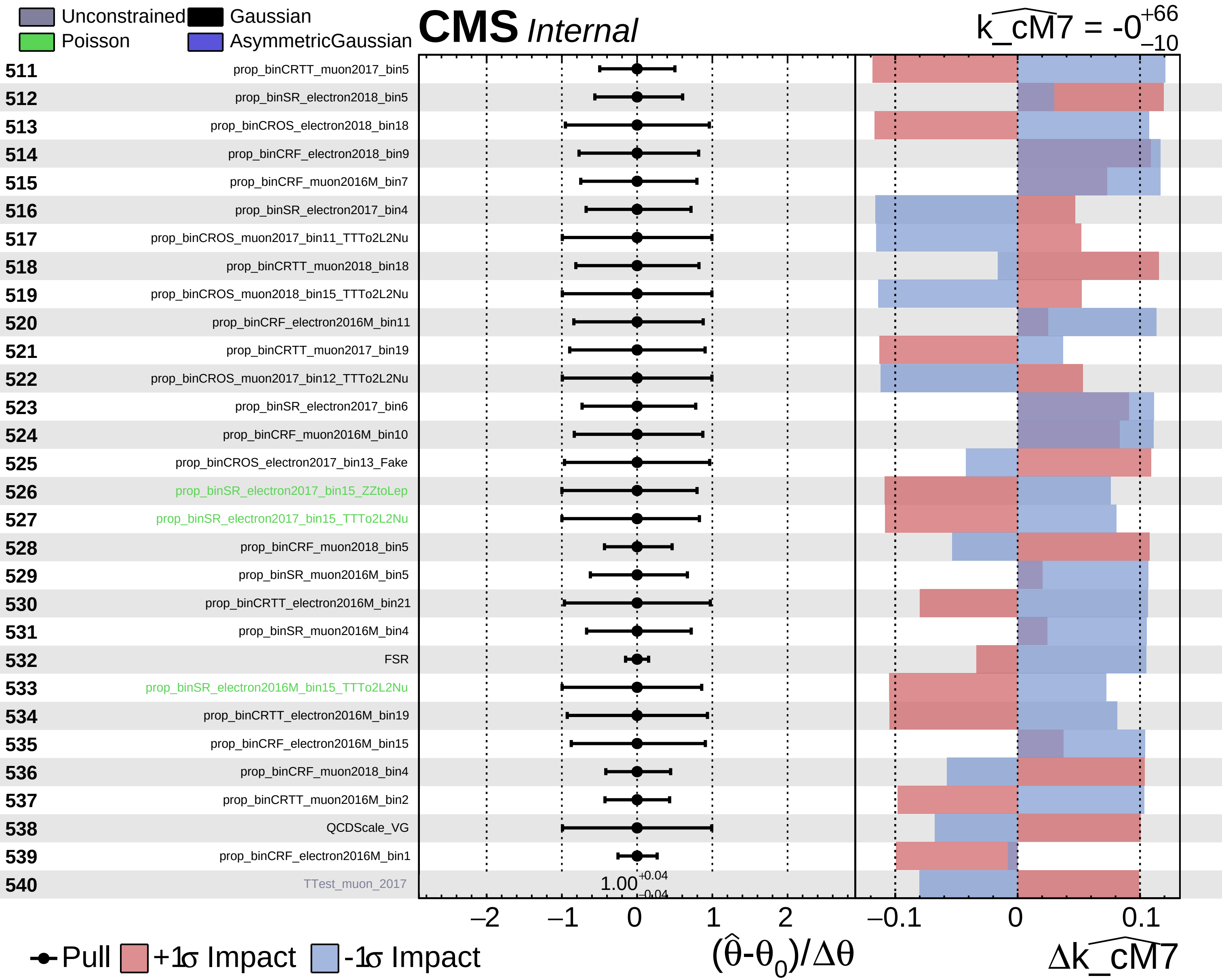
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM7 = -0 +66 −10
421 TTest_muon_2016M
1.00 +0.05 −0.05
422 prop_binCRF_electron2018_bin14
423 prop_binCRTT_muon2017_bin4
424 prop_binCROS_electron2018_bin14_TTTo2L2Nu
425 prop_binSR_electron2016M_bin11
426 prop_binCRF_muon2017_bin13
427 prop_binSR_electron2018_bin7
428 prop_binSR_electron2016M_bin13
429 prop_binSR_muon2017_bin15_sm
430 prop_binCROS_electron2018_bin2
431 prop_binCRF_electron2016M_bin12
432 prop_binSR_electron2016M_bin12
433 prop_binCROS_muon2016M_bin4
434 prop_binSR_muon2016M_bin6
435 prop_binCROS_electron2016M_bin12_Fake
436 prop_binSR_electron2017_bin20_Triboson
437 prop_binCRF_electron2017_bin8
438 prop_binCRF_electron2017_bin13
439 prop_binSR_electron2018_bin12
440 prop_binCROS_muon2018_bin16_TTTo2L2Nu
441 QCDScale_TVX
442 prop_binCROS_muon2018_bin21_TTTo2L2Nu
443 prop_binSR_electron2017_bin15_TVX
444 prop_binCRTT_muon2017_bin2
445 prop_binSR_muon2016M_bin19_TTTo2L2Nu
446 prop_binCROS_electron2017_bin11_TTTo2L2Nu
447 prop_binSR_electron2017_bin20_TTTo2L2Nu
448 prop_binCRF_muon2017_bin10
449 prop_binSR_muon2016M_bin8
450 prop_binCROS_muon2018_bin14_Fake
−2 −1 0 1 2
−0.2 0 0.2
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δk_cM7



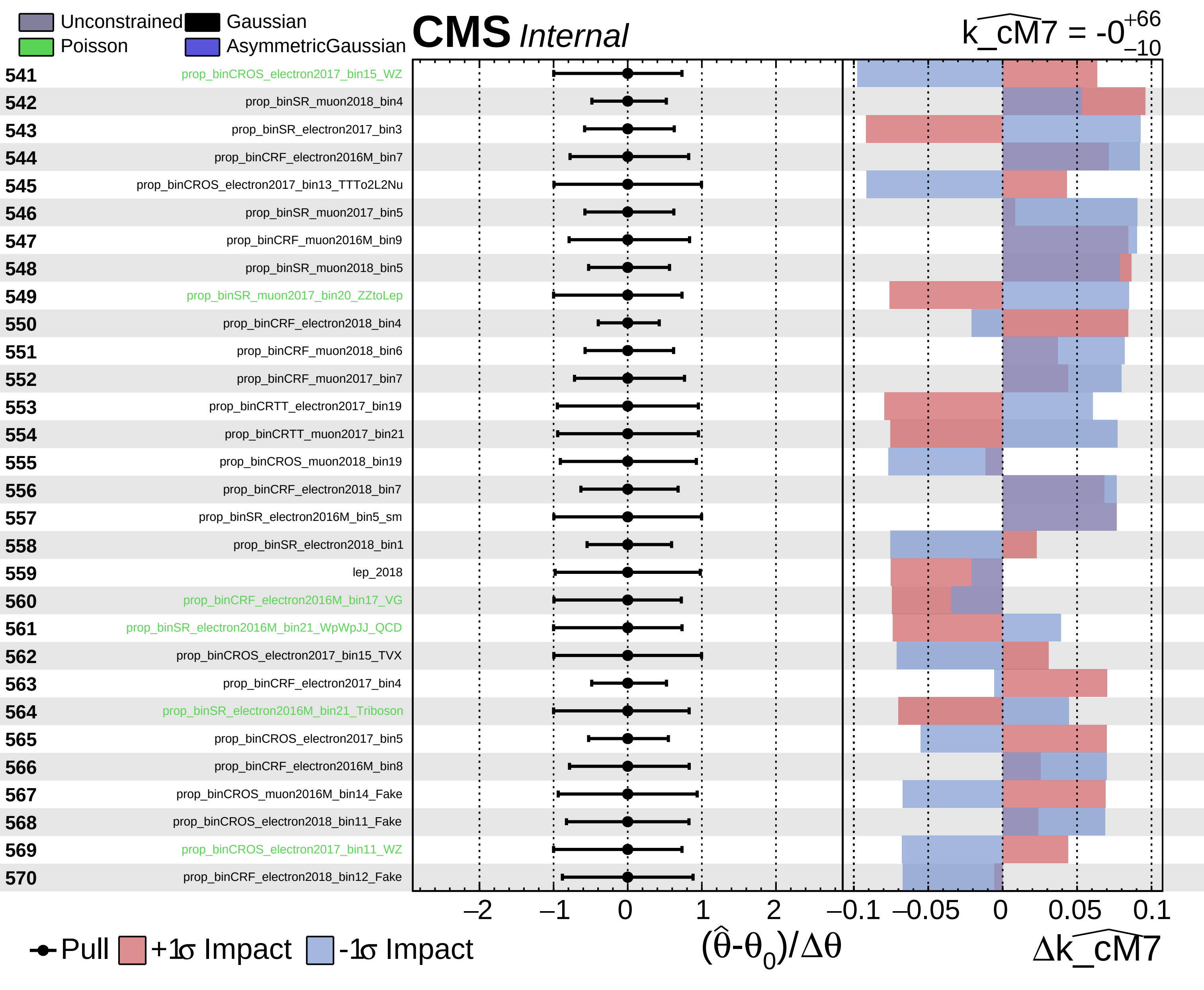
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM7 = -0 +66 −10
451 prop_binSR_electron2017_bin7
452 lep_2017
453 prop_binCRF_electron2018_bin15
454 prop_binCRF_muon2016M_bin12
455 prop_binCRF_electron2018_bin2
456 prop_binSR_muon2018_bin8
457 prop_binCRF_electron2016M_bin14_WZ
458 prop_binCRTT_electron2018_bin19
459 prop_binCRTT_electron2017_bin2
460 prop_binCRF_muon2016M_bin14_Fake
461 prop_binCRF_electron2016M_bin13
462 prop_binCRF_electron2018_bin11
463 prop_binSR_muon2017_bin7
464 prop_binCRTT_muon2017_bin3
465 prop_binCROS_electron2018_bin6
466 prop_binCRF_electron2016M_bin9
467 prop_binCROS_electron2018_bin11_WrongSign
468 prop_binCRF_electron2017_bin10
469 prop_binCRF_electron2017_bin0
470 prop_binCRTT_muon2016M_bin19
471 prop_binCRF_muon2018_bin10
472 prop_binCRTT_muon2016M_bin3
473 prop_binSR_electron2016M_bin7
474 prop_binCRF_muon2016M_bin15
475 prop_binCRF_muon2018_bin12
476 prop_binSR_muon2016M_bin7
477 prop_binCRF_electron2018_bin8
478 prop_binSR_electron2018_bin6
479 prop_binCRF_muon2016M_bin13
480 prop_binSR_muon2018_bin3
−2 −1 0 1 2
−0.2 −0.1 0 0.1 0.2
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cM7



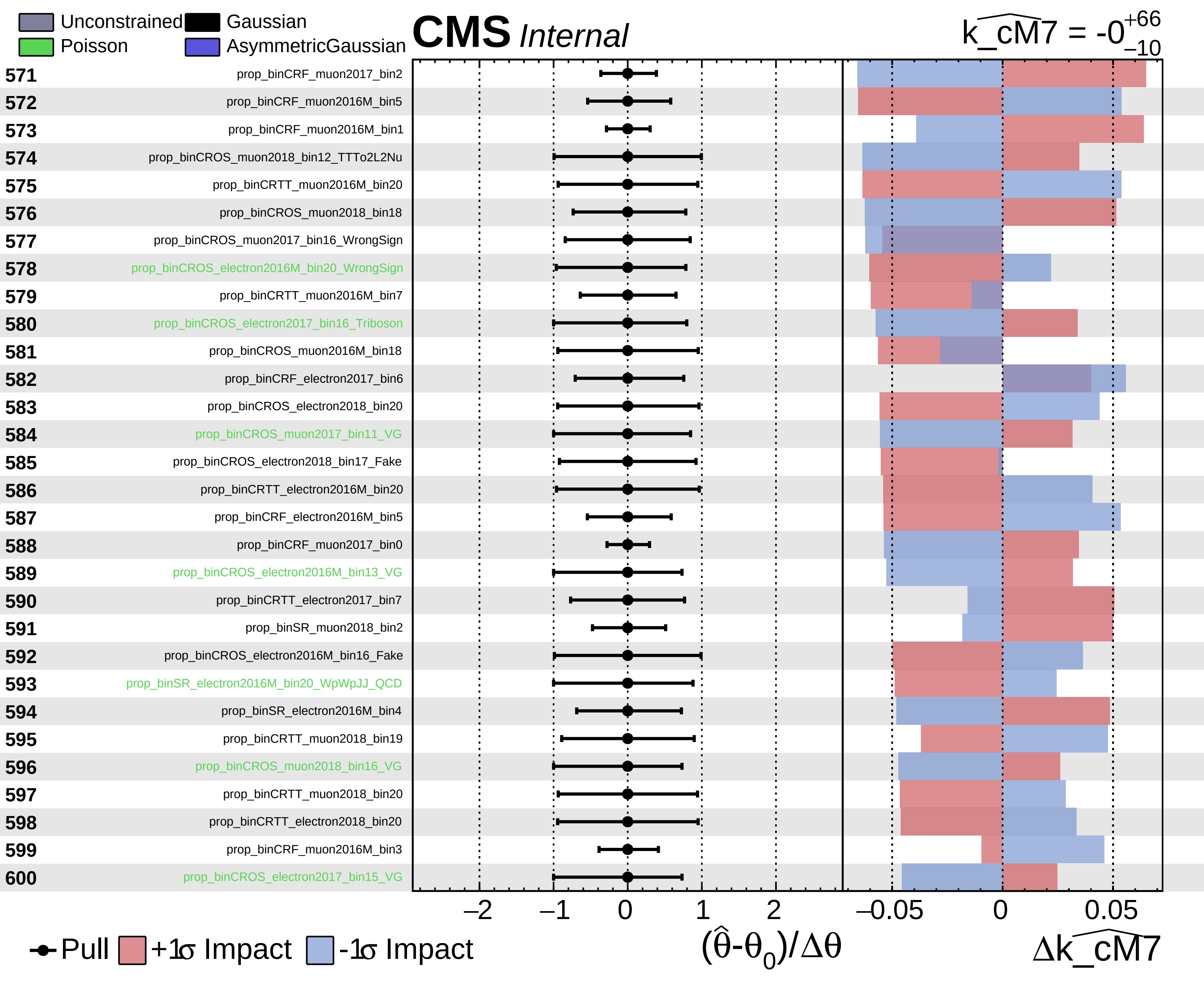
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM7 = -0 +66 −10
481 prop_binSR_muon2017_bin6
482 prop_binCRF_electron2017_bin1
483 prop_binCROS_muon2017_bin13_TTTo2L2Nu
484 prop_binCRF_electron2018_bin3
485 prop_binCRF_electron2017_bin9
486 prop_binCRF_electron2018_bin10
487 prop_binCROS_muon2016M_bin7_WrongSign
488 prop_binCROS_muon2016M_bin15_TTTo2L2Nu
489 prop_binSR_electron2018_bin8
490 prop_binCRF_muon2018_bin3
491 prop_binSR_muon2016M_bin9
492 prop_binCRF_electron2017_bin15
493 prop_binCROS_electron2016M_bin11_TTTo2L2Nu
494 prop_binCRF_muon2018_bin8
495 prop_binSR_muon2017_bin3
496 prop_binCRF_electron2018_bin12_sm
497 prop_binCRF_electron2016M_bin10
498 prop_binSR_electron2016M_bin5_Fake
499 prop_binCRF_muon2017_bin3
500 prop_binCRF_muon2016M_bin18
501 prop_binCRF_muon2018_bin7
502 prop_binCRF_muon2018_bin2
503 prop_binCRTT_muon2016M_bin6
504 prop_binCRF_muon2017_bin1
505 prop_binSR_electron2016M_bin15_VG
506 prop_binSR_electron2016M_bin5_WrongSign
507 prop_binSR_electron2016M_bin6
508 prop_binCROS_muon2016M_bin8_TTTo2L2Nu
509 prop_binSR_electron2018_bin21_TVX
510 prop_binCRTT_muon2016M_bin1
−2 −1 0 1 2
−0.1 0 0.1
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cM7

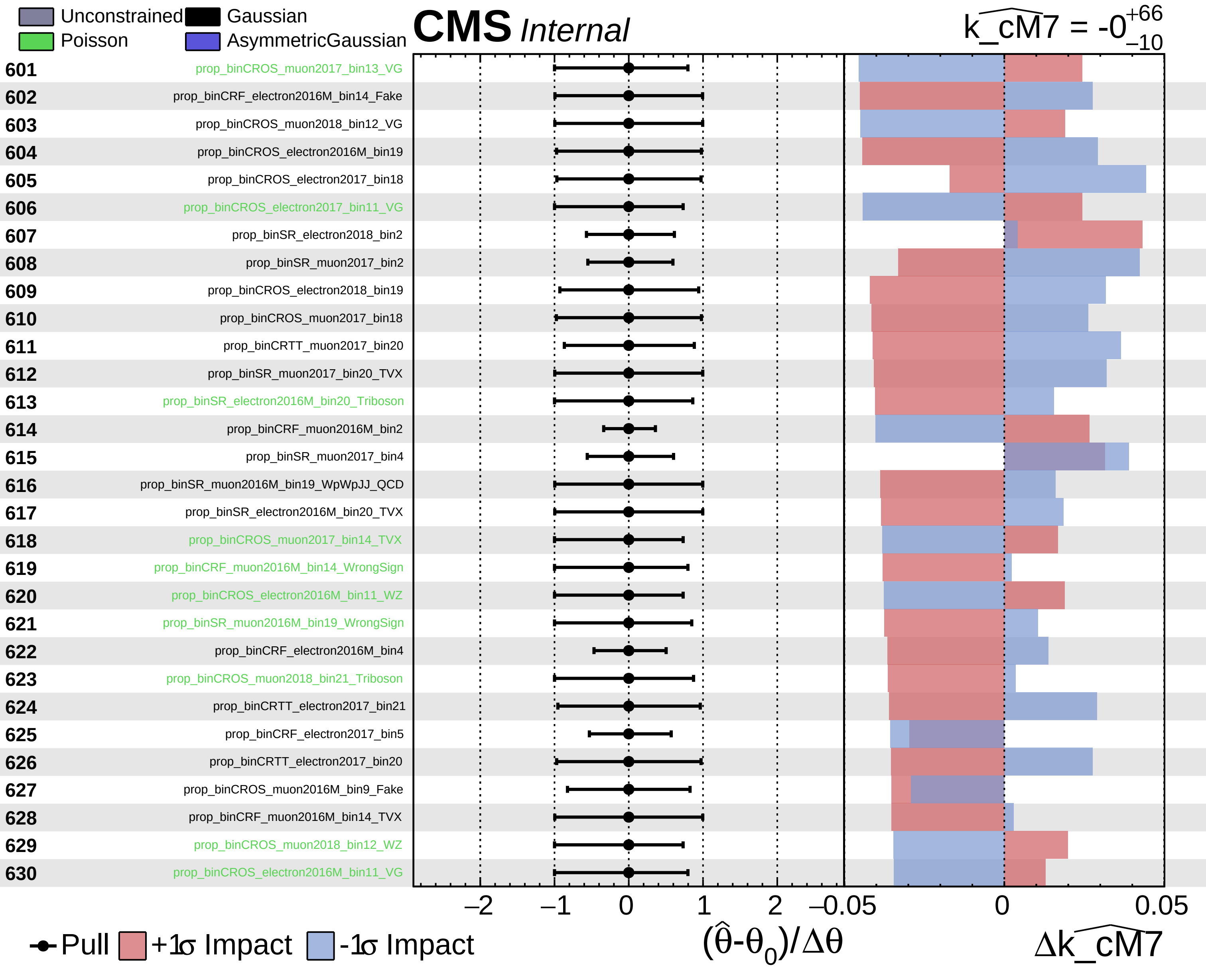


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM7 = -0 +66 −10
511 prop_binCRTT_muon2017_bin5
512 prop_binSR_electron2018_bin5
513 prop_binCROS_electron2018_bin18
514 prop_binCRF_electron2018_bin9
515 prop_binCRF_muon2016M_bin7
516 prop_binSR_electron2017_bin4
517 prop_binCROS_muon2017_bin11_TTTo2L2Nu
518 prop_binCRTT_muon2018_bin18
519 prop_binCROS_muon2018_bin15_TTTo2L2Nu
520 prop_binCRF_electron2016M_bin11
521 prop_binCRTT_muon2017_bin19
522 prop_binCROS_muon2017_bin12_TTTo2L2Nu
523 prop_binSR_electron2017_bin6
524 prop_binCRF_muon2016M_bin10
525 prop_binCROS_electron2017_bin13_Fake
526 prop_binSR_electron2017_bin15_ZZtoLep
527 prop_binSR_electron2017_bin15_TTTo2L2Nu
528 prop_binCRF_muon2018_bin5
529 prop_binSR_muon2016M_bin5
530 prop_binCRTT_electron2016M_bin21
531 prop_binSR_muon2016M_bin4
532 FSR
533 prop_binSR_electron2016M_bin15_TTTo2L2Nu
534 prop_binCRTT_electron2016M_bin19
535 prop_binCRF_electron2016M_bin15
536 prop_binCRF_muon2018_bin4
537 prop_binCRTT_muon2016M_bin2
538 QCDScale_VG
539 prop_binCRF_electron2016M_bin1
540 TTest_muon_2017
1.00 +0.04 −0.04
−2 −1 0 1 2
−0.1 0 0.1
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cM7

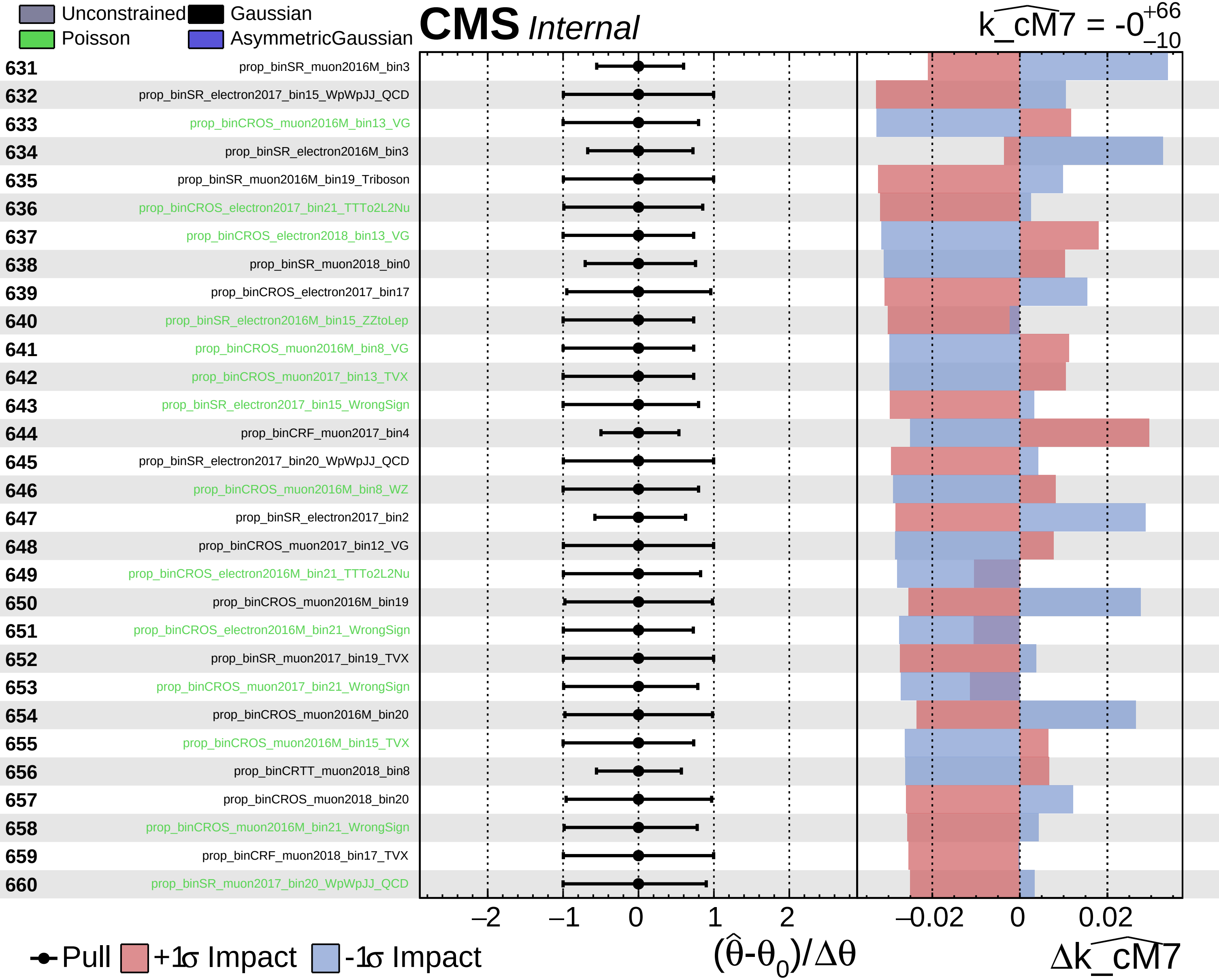


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM7 = -0 +66 −10
541 prop_binCROS_electron2017_bin15_WZ
542 prop_binSR_muon2018_bin4
543 prop_binSR_electron2017_bin3
544 prop_binCRF_electron2016M_bin7
545 prop_binCROS_electron2017_bin13_TTTo2L2Nu
546 prop_binSR_muon2017_bin5
547 prop_binCRF_muon2016M_bin9
548 prop_binSR_muon2018_bin5
549 prop_binSR_muon2017_bin20_ZZtoLep
550 prop_binCRF_electron2018_bin4
551 prop_binCRF_muon2018_bin6
552 prop_binCRF_muon2017_bin7
553 prop_binCRTT_electron2017_bin19
554 prop_binCRTT_muon2017_bin21
555 prop_binCROS_muon2018_bin19
556 prop_binCRF_electron2018_bin7
557 prop_binSR_electron2016M_bin5_sm
558 prop_binSR_electron2018_bin1
559 lep_2018
560 prop_binCRF_electron2016M_bin17_VG
561 prop_binSR_electron2016M_bin21_WpWpJJ_QCD
562 prop_binCROS_electron2017_bin15_TVX
563 prop_binCRF_electron2017_bin4
564 prop_binSR_electron2016M_bin21_Triboson
565 prop_binCROS_electron2017_bin5
566 prop_binCRF_electron2016M_bin8
567 prop_binCROS_muon2016M_bin14_Fake
568 prop_binCROS_electron2018_bin11_Fake
569 prop_binCROS_electron2017_bin11_WZ
570 prop_binCRF_electron2018_bin12_Fake
−2 −1 0 1 2
−0.1 −0.05 0 0.05 0.1
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cM7

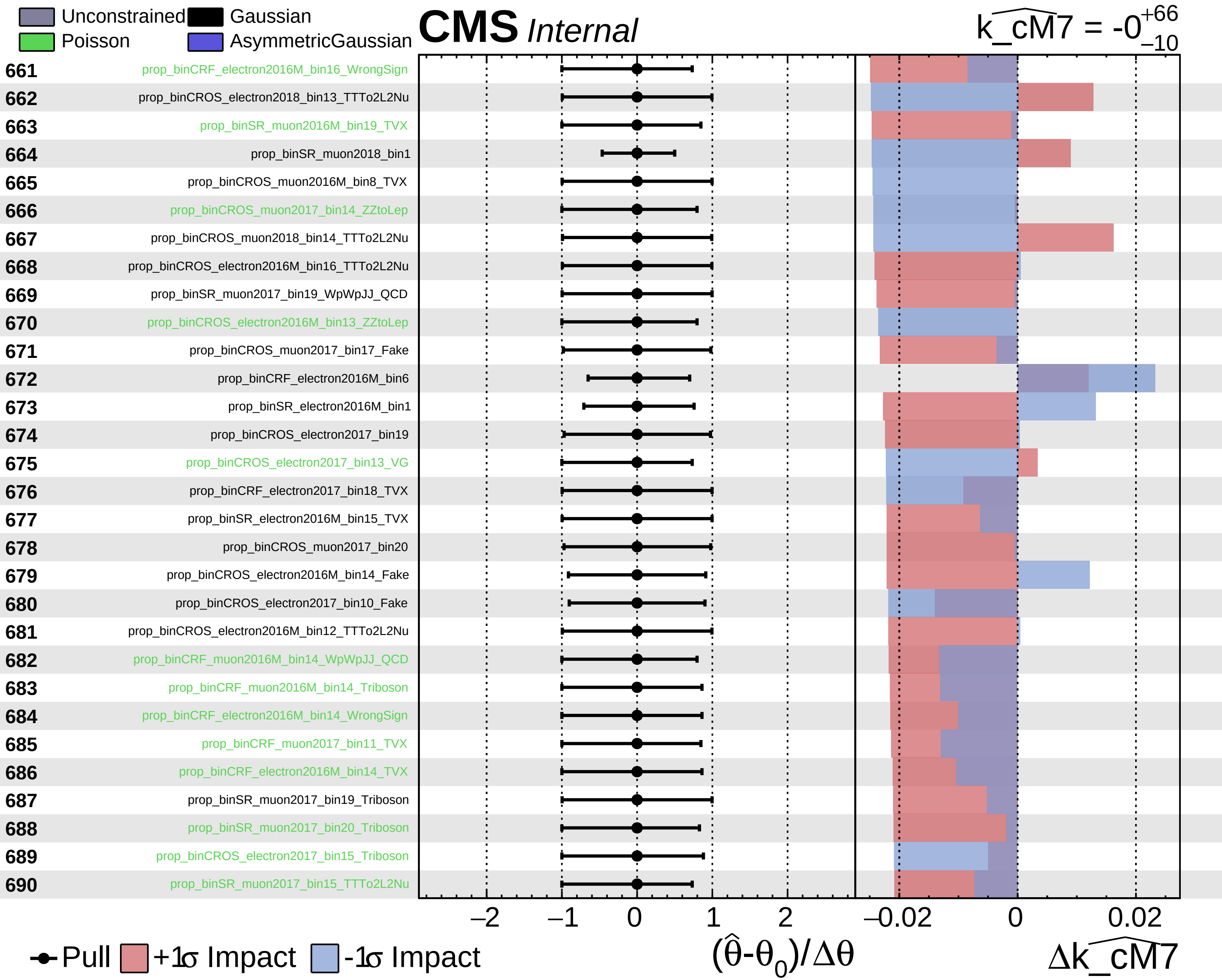

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM7 = -0 +66 −10
571 prop_binCRF_muon2017_bin2
572 prop_binCRF_muon2016M_bin5
573 prop_binCRF_muon2016M_bin1
574 prop_binCROS_muon2018_bin12_TTTo2L2Nu
575 prop_binCRTT_muon2016M_bin20
576 prop_binCROS_muon2018_bin18
577 prop_binCROS_muon2017_bin16_WrongSign
578 prop_binCROS_electron2016M_bin20_WrongSign
579 prop_binCRTT_muon2016M_bin7
580 prop_binCROS_electron2017_bin16_Triboson
581 prop_binCROS_muon2016M_bin18
582 prop_binCRF_electron2017_bin6
583 prop_binCROS_electron2018_bin20
584 prop_binCROS_muon2017_bin11_VG
585 prop_binCROS_electron2018_bin17_Fake
586 prop_binCRTT_electron2016M_bin20
587 prop_binCRF_electron2016M_bin5
588 prop_binCRF_muon2017_bin0
589 prop_binCROS_electron2016M_bin13_VG
590 prop_binCRTT_electron2017_bin7
591 prop_binSR_muon2018_bin2
592 prop_binCROS_electron2016M_bin16_Fake
593 prop_binSR_electron2016M_bin20_WpWpJJ_QCD
594 prop_binSR_electron2016M_bin4
595 prop_binCRTT_muon2018_bin19
596 prop_binCROS_muon2018_bin16_VG
597 prop_binCRTT_muon2018_bin20
598 prop_binCRTT_electron2018_bin20
599 prop_binCRF_muon2016M_bin3
600 prop_binCROS_electron2017_bin15_VG
−2 −1 0 1 2
−0.05 0 0.05
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cM7



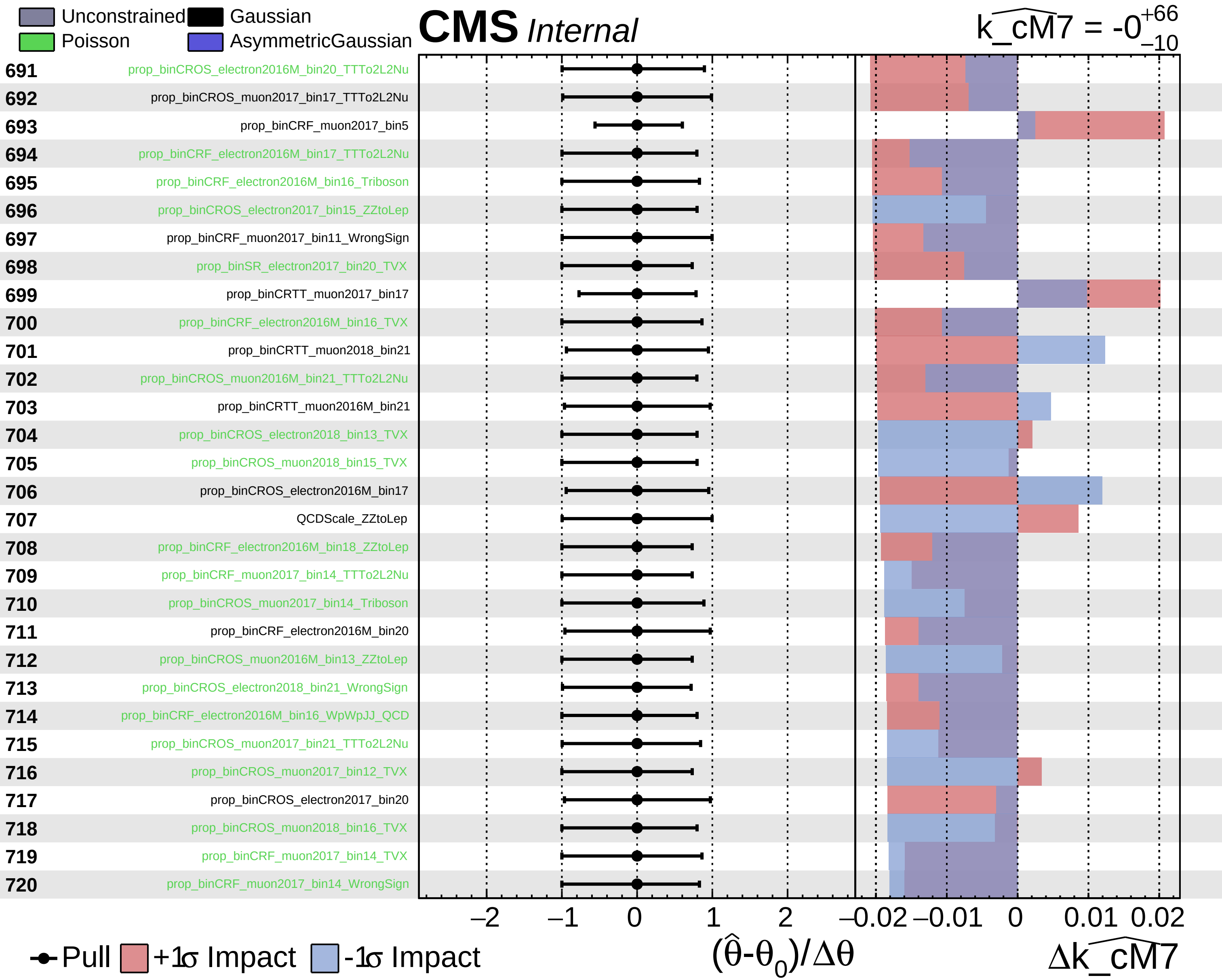
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM7 = -0 +66 −10
601 prop_binCROS_muon2017_bin13_VG
602 prop_binCRF_electron2016M_bin14_Fake
603 prop_binCROS_muon2018_bin12_VG
604 prop_binCROS_electron2016M_bin19
605 prop_binCROS_electron2017_bin18
606 prop_binCROS_electron2017_bin11_VG
607 prop_binSR_electron2018_bin2
608 prop_binSR_muon2017_bin2
609 prop_binCROS_electron2018_bin19
610 prop_binCROS_muon2017_bin18
611 prop_binCRTT_muon2017_bin20
612 prop_binSR_muon2017_bin20_TVX
613 prop_binSR_electron2016M_bin20_Triboson
614 prop_binCRF_muon2016M_bin2
615 prop_binSR_muon2017_bin4
616 prop_binSR_muon2016M_bin19_WpWpJJ_QCD
617 prop_binSR_electron2016M_bin20_TVX
618 prop_binCROS_muon2017_bin14_TVX
619 prop_binCRF_muon2016M_bin14_WrongSign
620 prop_binCROS_electron2016M_bin11_WZ
621 prop_binSR_muon2016M_bin19_WrongSign
622 prop_binCRF_electron2016M_bin4
623 prop_binCROS_muon2018_bin21_Triboson
624 prop_binCRTT_electron2017_bin21
625 prop_binCRF_electron2017_bin5
626 prop_binCRTT_electron2017_bin20
627 prop_binCROS_muon2016M_bin9_Fake
628 prop_binCRF_muon2016M_bin14_TVX
629 prop_binCROS_muon2018_bin12_WZ
630 prop_binCROS_electron2016M_bin11_VG
−2 −1 0 1 2
−0.05 0 0.05
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cM7

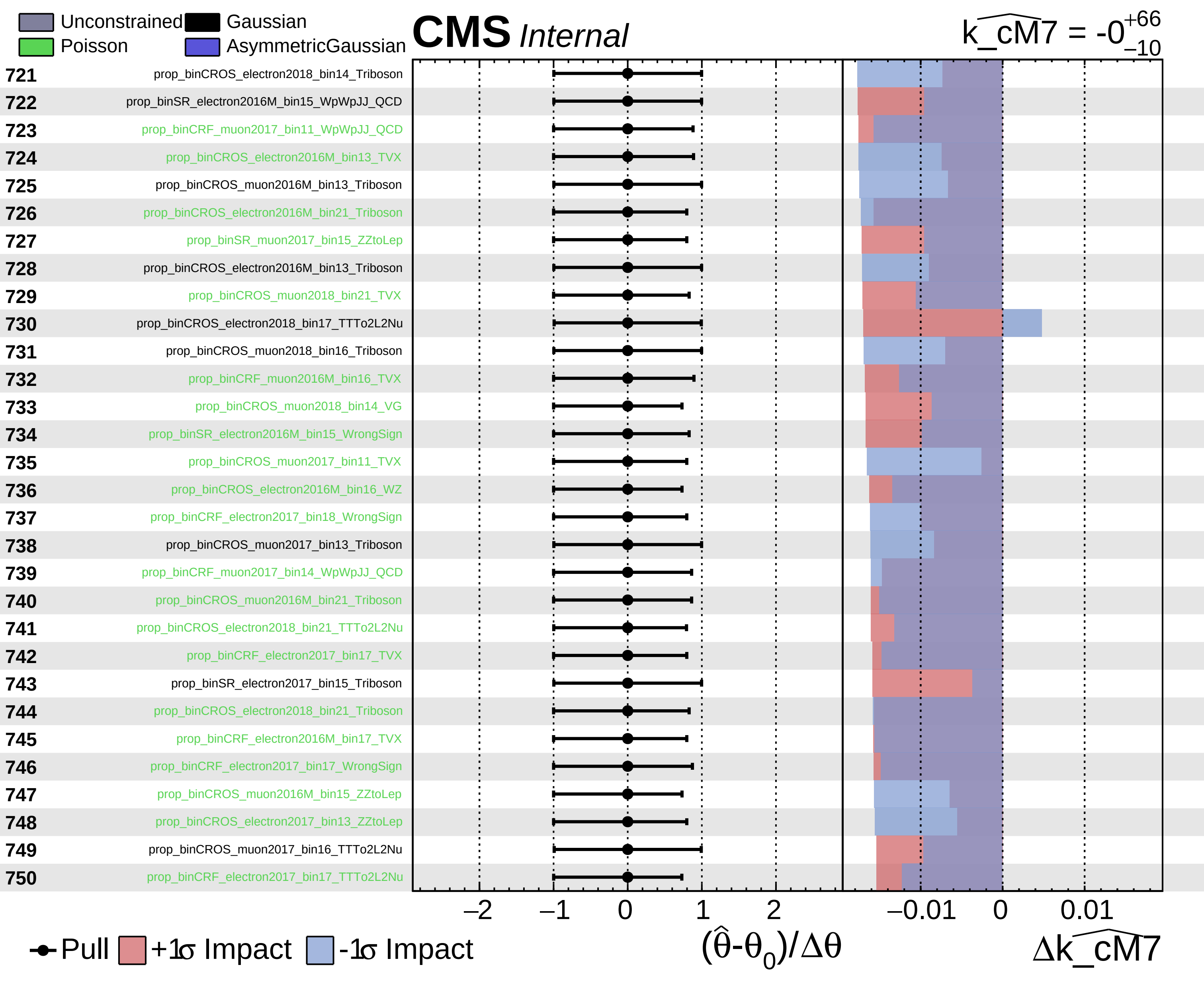


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM7 = -0 +66 −10
631 prop_binSR_muon2016M_bin3
632 prop_binSR_electron2017_bin15_WpWpJJ_QCD
633 prop_binCROS_muon2016M_bin13_VG
634 prop_binSR_electron2016M_bin3
635 prop_binSR_muon2016M_bin19_Triboson
636 prop_binCROS_electron2017_bin21_TTTo2L2Nu
637 prop_binCROS_electron2018_bin13_VG
638 prop_binSR_muon2018_bin0
639 prop_binCROS_electron2017_bin17
640 prop_binSR_electron2016M_bin15_ZZtoLep
641 prop_binCROS_muon2016M_bin8_VG
642 prop_binCROS_muon2017_bin13_TVX
643 prop_binSR_electron2017_bin15_WrongSign
644 prop_binCRF_muon2017_bin4
645 prop_binSR_electron2017_bin20_WpWpJJ_QCD
646 prop_binCROS_muon2016M_bin8_WZ
647 prop_binSR_electron2017_bin2
648 prop_binCROS_muon2017_bin12_VG
649 prop_binCROS_electron2016M_bin21_TTTo2L2Nu
650 prop_binCROS_muon2016M_bin19
651 prop_binCROS_electron2016M_bin21_WrongSign
652 prop_binSR_muon2017_bin19_TVX
653 prop_binCROS_muon2017_bin21_WrongSign
654 prop_binCROS_muon2016M_bin20
655 prop_binCROS_muon2016M_bin15_TVX
656 prop_binCRTT_muon2018_bin8
657 prop_binCROS_muon2018_bin20
658 prop_binCROS_muon2016M_bin21_WrongSign
659 prop_binCRF_muon2018_bin17_TVX
660 prop_binSR_muon2017_bin20_WpWpJJ_QCD
−2
−1
0
1
2
−0.02
0
0.02
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cM7

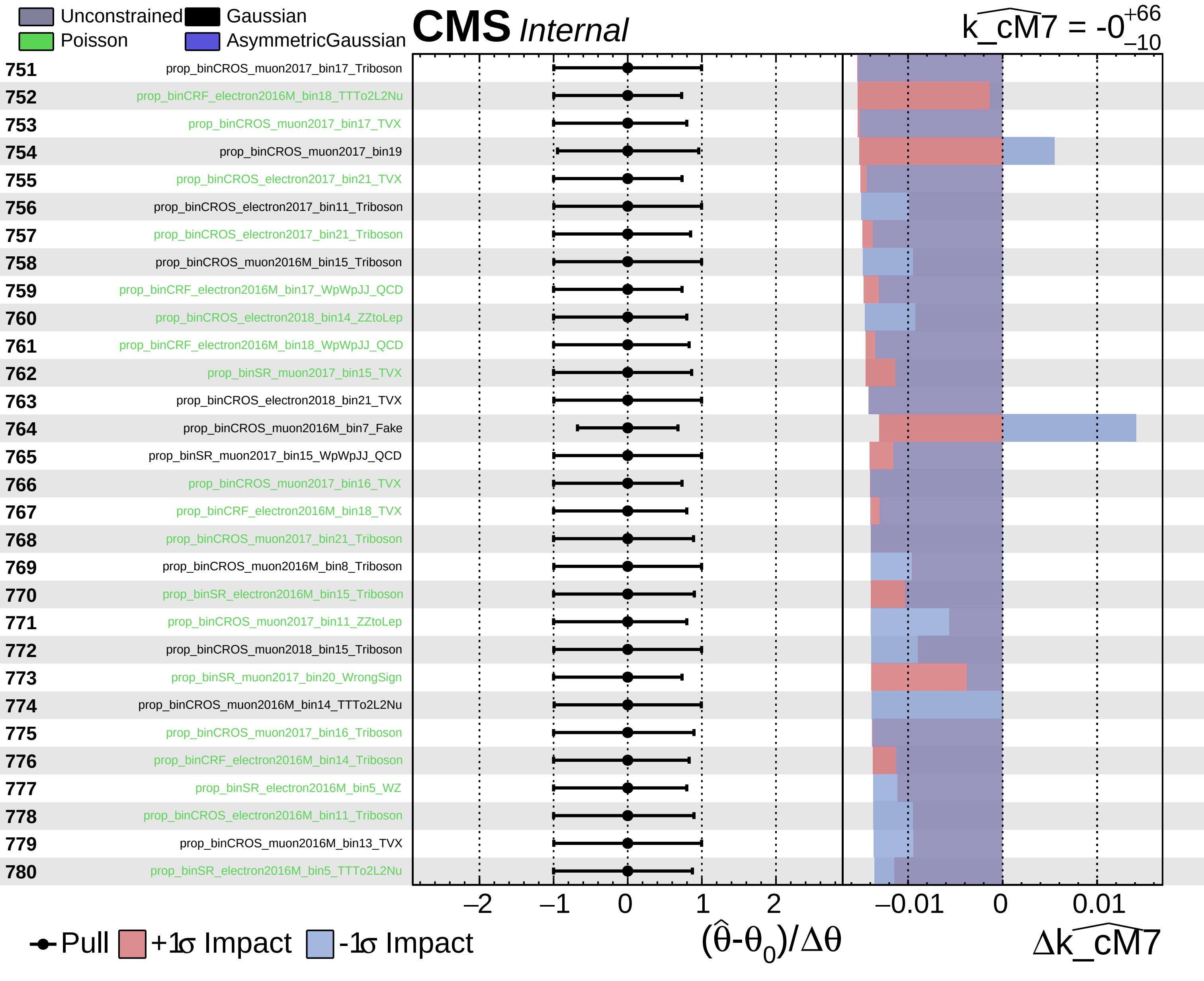


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM7 = -0 +66 −10
661 prop_binCRF_electron2016M_bin16_WrongSign
662 prop_binCROS_electron2018_bin13_TTTo2L2Nu
663 prop_binSR_muon2016M_bin19_TVX
664 prop_binSR_muon2018_bin1
665 prop_binCROS_muon2016M_bin8_TVX
666 prop_binCROS_muon2017_bin14_ZZtoLep
667 prop_binCROS_muon2018_bin14_TTTo2L2Nu
668 prop_binCROS_electron2016M_bin16_TTTo2L2Nu
669 prop_binSR_muon2017_bin19_WpWpJJ_QCD
670 prop_binCROS_electron2016M_bin13_ZZtoLep
671 prop_binCROS_muon2017_bin17_Fake
672 prop_binCRF_electron2016M_bin6
673 prop_binSR_electron2016M_bin1
674 prop_binCROS_electron2017_bin19
675 prop_binCROS_electron2017_bin13_VG
676 prop_binCRF_electron2017_bin18_TVX
677 prop_binSR_electron2016M_bin15_TVX
678 prop_binCROS_muon2017_bin20
679 prop_binCROS_electron2016M_bin14_Fake
680 prop_binCROS_electron2017_bin10_Fake
681 prop_binCROS_electron2016M_bin12_TTTo2L2Nu
682 prop_binCRF_muon2016M_bin14_WpWpJJ_QCD
683 prop_binCRF_muon2016M_bin14_Triboson
684 prop_binCRF_electron2016M_bin14_WrongSign
685 prop_binCRF_muon2017_bin11_TVX
686 prop_binCRF_electron2016M_bin14_TVX
687 prop_binSR_muon2017_bin19_Triboson
688 prop_binSR_muon2017_bin20_Triboson
689 prop_binCROS_electron2017_bin15_Triboson
690 prop_binSR_muon2017_bin15_TTTo2L2Nu
−2
−1
0
1
2
−0.02
0
0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cM7

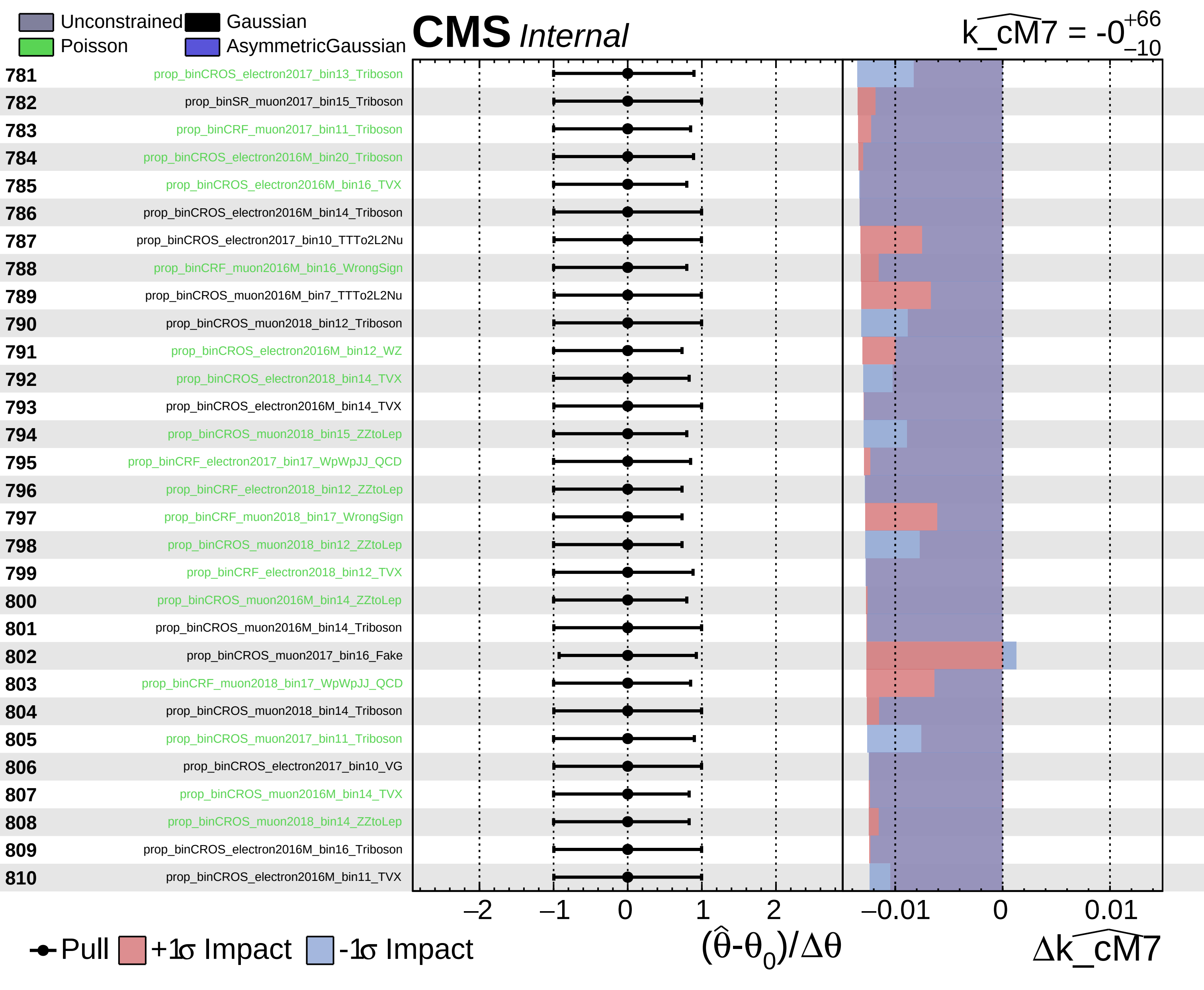

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM7} = -0^{+66}_{-10}$
691 prop_binCROS_electron2016M_bin20_TTTo2L2Nu
692 prop_binCROS_muon2017_bin17_TTTo2L2Nu
693 prop_binCRF_muon2017_bin5
694 prop_binCRF_electron2016M_bin17_TTTo2L2Nu
695 prop_binCRF_electron2016M_bin16_Triboson
696 prop_binCROS_electron2017_bin15_ZZtoLep
697 prop_binCRF_muon2017_bin11_WrongSign
698 prop_binSR_electron2017_bin20_TVX
699 prop_binCRTT_muon2017_bin17
700 prop_binCRF_electron2016M_bin16_TVX
701 prop_binCRTT_muon2018_bin21
702 prop_binCROS_muon2016M_bin21_TTTo2L2Nu
703 prop_binCRTT_muon2016M_bin21
704 prop_binCROS_electron2018_bin13_TVX
705 prop_binCROS_muon2018_bin15_TVX
706 prop_binCROS_electron2016M_bin17
707 QCDScale_ZZtoLep
708 prop_binCRF_electron2016M_bin18_ZZtoLep
709 prop_binCRF_muon2017_bin14_TTTo2L2Nu
710 prop_binCROS_muon2017_bin14_Triboson
711 prop_binCRF_electron2016M_bin20
712 prop_binCROS_muon2016M_bin13_ZZtoLep
713 prop_binCROS_electron2018_bin21_WrongSign
714 prop_binCRF_electron2016M_bin16_WpWpJJ_QCD
715 prop_binCROS_muon2017_bin21_TTTo2L2Nu
716 prop_binCROS_muon2017_bin12_TVX
717 prop_binCROS_electron2017_bin20
718 prop_binCROS_muon2018_bin16_TVX
719 prop_binCRF_muon2017_bin14_TVX
720 prop_binCRF_muon2017_bin14_WrongSign
−2 −1 0 1 2
−0.02 −0.01 0 0.01 0.02
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM7}$



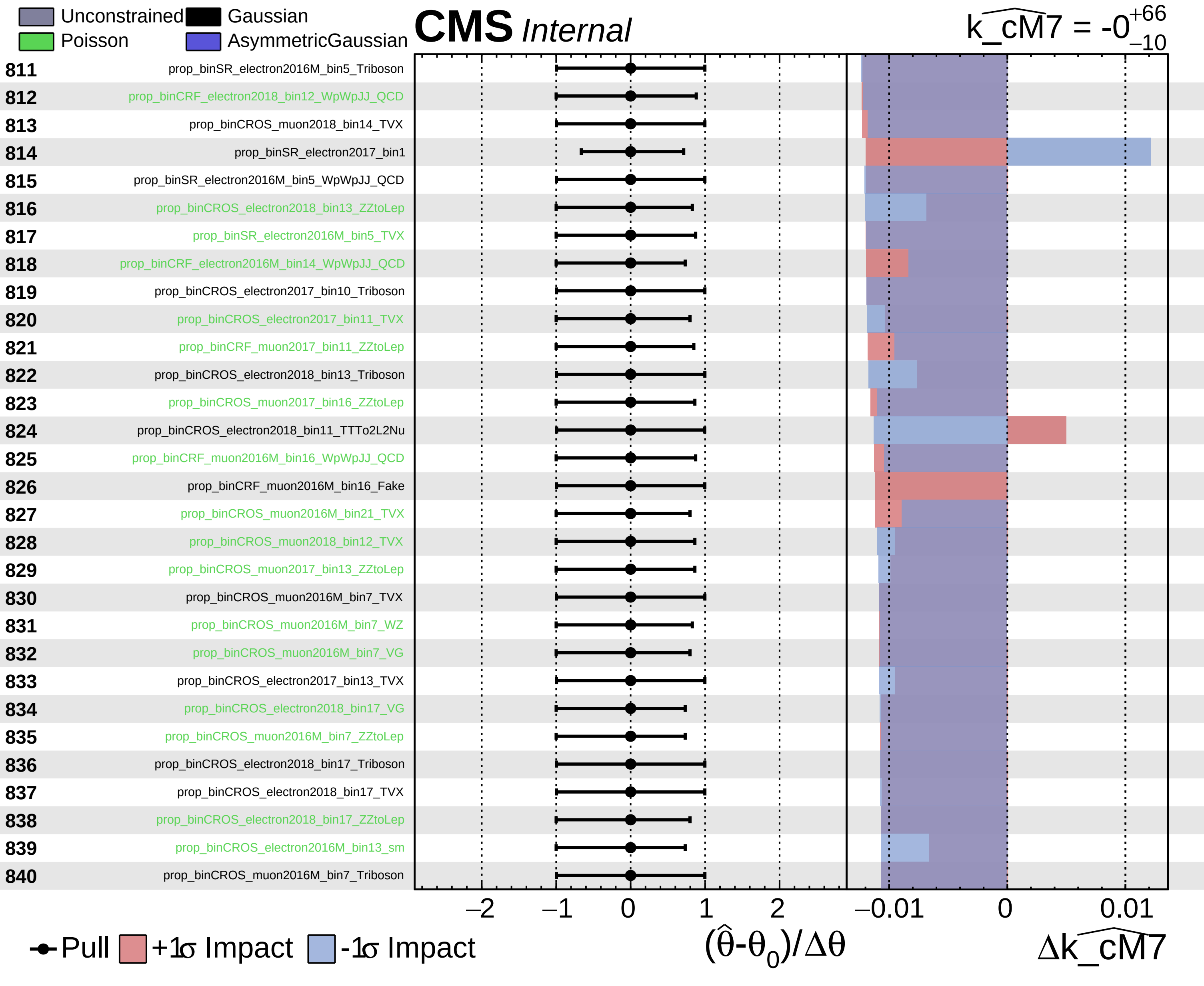
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM7} = -0^{+66}_{-10}$
721 prop_binCROS_electron2018_bin14_Triboson
722 prop_binSR_electron2016M_bin15_WpWpJJ_QCD
723 prop_binCRF_muon2017_bin11_WpWpJJ_QCD
724 prop_binCROS_electron2016M_bin13_TVX
725 prop_binCROS_muon2016M_bin13_Triboson
726 prop_binCROS_electron2016M_bin21_Triboson
727 prop_binSR_muon2017_bin15_ZZtoLep
728 prop_binCROS_electron2016M_bin13_Triboson
729 prop_binCROS_muon2018_bin21_TVX
730 prop_binCROS_electron2018_bin17_TTTo2L2Nu
731 prop_binCROS_muon2018_bin16_Triboson
732 prop_binCRF_muon2016M_bin16_TVX
733 prop_binCROS_muon2018_bin14_VG
734 prop_binSR_electron2016M_bin15_WrongSign
735 prop_binCROS_muon2017_bin11_TVX
736 prop_binCROS_electron2016M_bin16_WZ
737 prop_binCRF_electron2017_bin18_WrongSign
738 prop_binCROS_muon2017_bin13_Triboson
739 prop_binCRF_muon2017_bin14_WpWpJJ_QCD
740 prop_binCROS_muon2016M_bin21_Triboson
741 prop_binCROS_electron2018_bin21_TTTo2L2Nu
742 prop_binCRF_electron2017_bin17_TVX
743 prop_binSR_electron2017_bin15_Triboson
744 prop_binCROS_electron2018_bin21_Triboson
745 prop_binCRF_electron2016M_bin17_TVX
746 prop_binCRF_electron2017_bin17_WrongSign
747 prop_binCROS_muon2016M_bin15_ZZtoLep
748 prop_binCROS_electron2017_bin13_ZZtoLep
749 prop_binCROS_muon2017_bin16_TTTo2L2Nu
750 prop_binCRF_electron2017_bin17_TTTo2L2Nu
−2
−1
0
1
2
−0.01
0
0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM7}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM7} = -0^{+66}_{-10}$
751 prop_binCROS_muon2017_bin17_Triboson
752 prop_binCRF_electron2016M_bin18_TTTo2L2Nu
753 prop_binCROS_muon2017_bin17_TVX
754 prop_binCROS_muon2017_bin19
755 prop_binCROS_electron2017_bin21_TVX
756 prop_binCROS_electron2017_bin11_Triboson
757 prop_binCROS_electron2017_bin21_Triboson
758 prop_binCROS_muon2016M_bin15_Triboson
759 prop_binCRF_electron2016M_bin17_WpWpJJ_QCD
760 prop_binCROS_electron2018_bin14_ZZtoLep
761 prop_binCRF_electron2016M_bin18_WpWpJJ_QCD
762 prop_binSR_muon2017_bin15_TVX
763 prop_binCROS_electron2018_bin21_TVX
764 prop_binCROS_muon2016M_bin7_Fake
765 prop_binSR_muon2017_bin15_WpWpJJ_QCD
766 prop_binCROS_muon2017_bin16_TVX
767 prop_binCRF_electron2016M_bin18_TVX
768 prop_binCROS_muon2017_bin21_Triboson
769 prop_binCROS_muon2016M_bin8_Triboson
770 prop_binSR_electron2016M_bin15_Triboson
771 prop_binCROS_muon2017_bin11_ZZtoLep
772 prop_binCROS_muon2018_bin15_Triboson
773 prop_binSR_muon2017_bin20_WrongSign
774 prop_binCROS_muon2016M_bin14_TTTo2L2Nu
775 prop_binCROS_muon2017_bin16_Triboson
776 prop_binCRF_electron2016M_bin14_Triboson
777 prop_binSR_electron2016M_bin5_WZ
778 prop_binCROS_electron2016M_bin11_Triboson
779 prop_binCROS_muon2016M_bin13_TVX
780 prop_binSR_electron2016M_bin5_TTTo2L2Nu
−2 −1 0 1 2
−0.01 0 0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM7}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM7 = -0 +66 −10
781 prop_binCROS_electron2017_bin13_Triboson
782 prop_binSR_muon2017_bin15_Triboson
783 prop_binCRF_muon2017_bin11_Triboson
784 prop_binCROS_electron2016M_bin20_Triboson
785 prop_binCROS_electron2016M_bin16_TVX
786 prop_binCROS_electron2016M_bin14_Triboson
787 prop_binCROS_electron2017_bin10_TTTo2L2Nu
788 prop_binCRF_muon2016M_bin16_WrongSign
789 prop_binCROS_muon2016M_bin7_TTTo2L2Nu
790 prop_binCROS_muon2018_bin12_Triboson
791 prop_binCROS_electron2016M_bin12_WZ
792 prop_binCROS_electron2018_bin14_TVX
793 prop_binCROS_electron2016M_bin14_TVX
794 prop_binCROS_muon2018_bin15_ZZtoLep
795 prop_binCRF_electron2017_bin17_WpWpJJ_QCD
796 prop_binCRF_electron2018_bin12_ZZtoLep
797 prop_binCRF_muon2018_bin17_WrongSign
798 prop_binCROS_muon2018_bin12_ZZtoLep
799 prop_binCRF_electron2018_bin12_TVX
800 prop_binCROS_muon2016M_bin14_ZZtoLep
801 prop_binCROS_muon2016M_bin14_Triboson
802 prop_binCROS_muon2017_bin16_Fake
803 prop_binCRF_muon2018_bin17_WpWpJJ_QCD
804 prop_binCROS_muon2018_bin14_Triboson
805 prop_binCROS_muon2017_bin11_Triboson
806 prop_binCROS_electron2017_bin10_VG
807 prop_binCROS_muon2016M_bin14_TVX
808 prop_binCROS_muon2018_bin14_ZZtoLep
809 prop_binCROS_electron2016M_bin16_Triboson
810 prop_binCROS_electron2016M_bin11_TVX
−2 −1 0 1 2
−0.01 0 0.01
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cM7


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM7 = -0+66 −10
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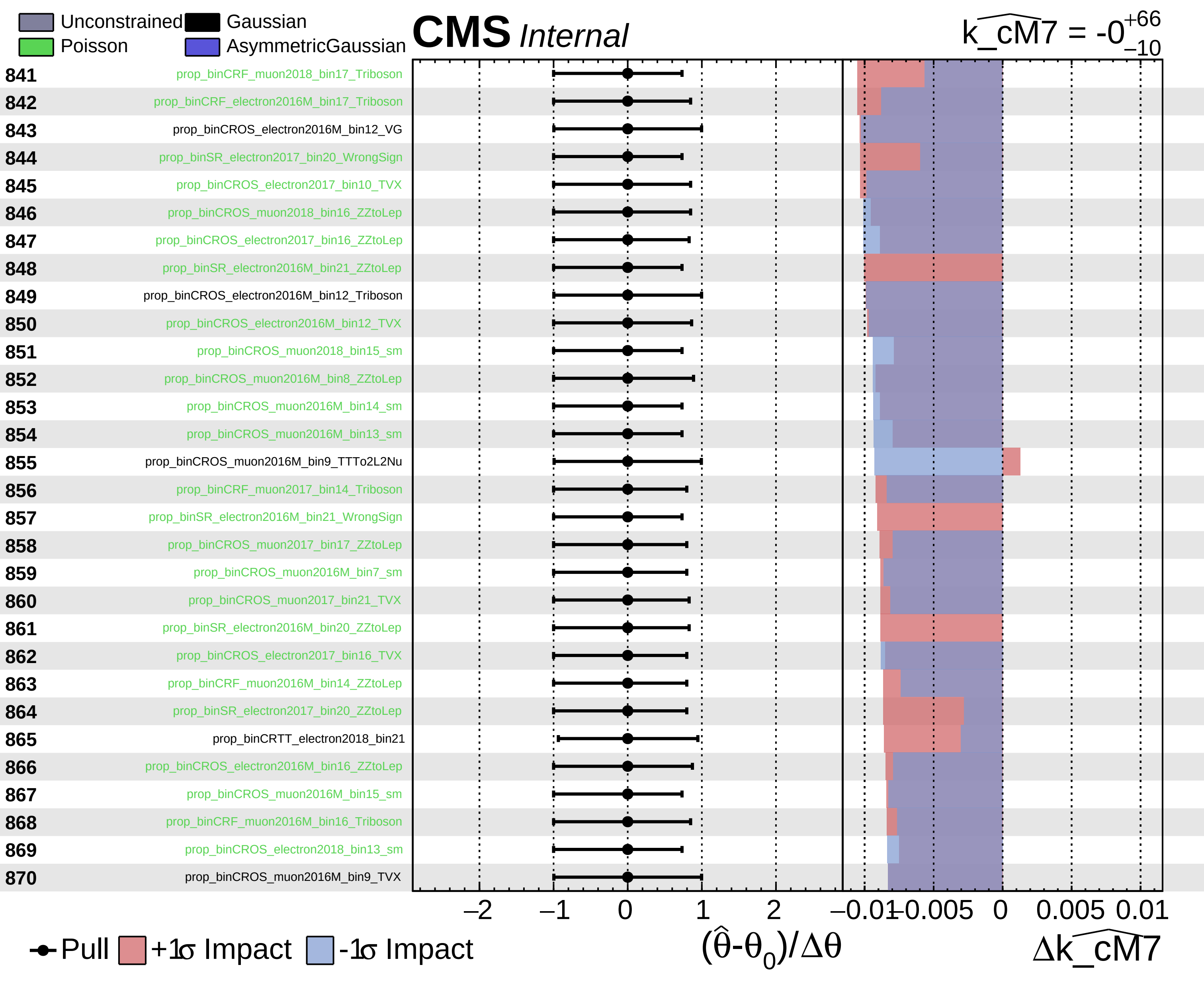
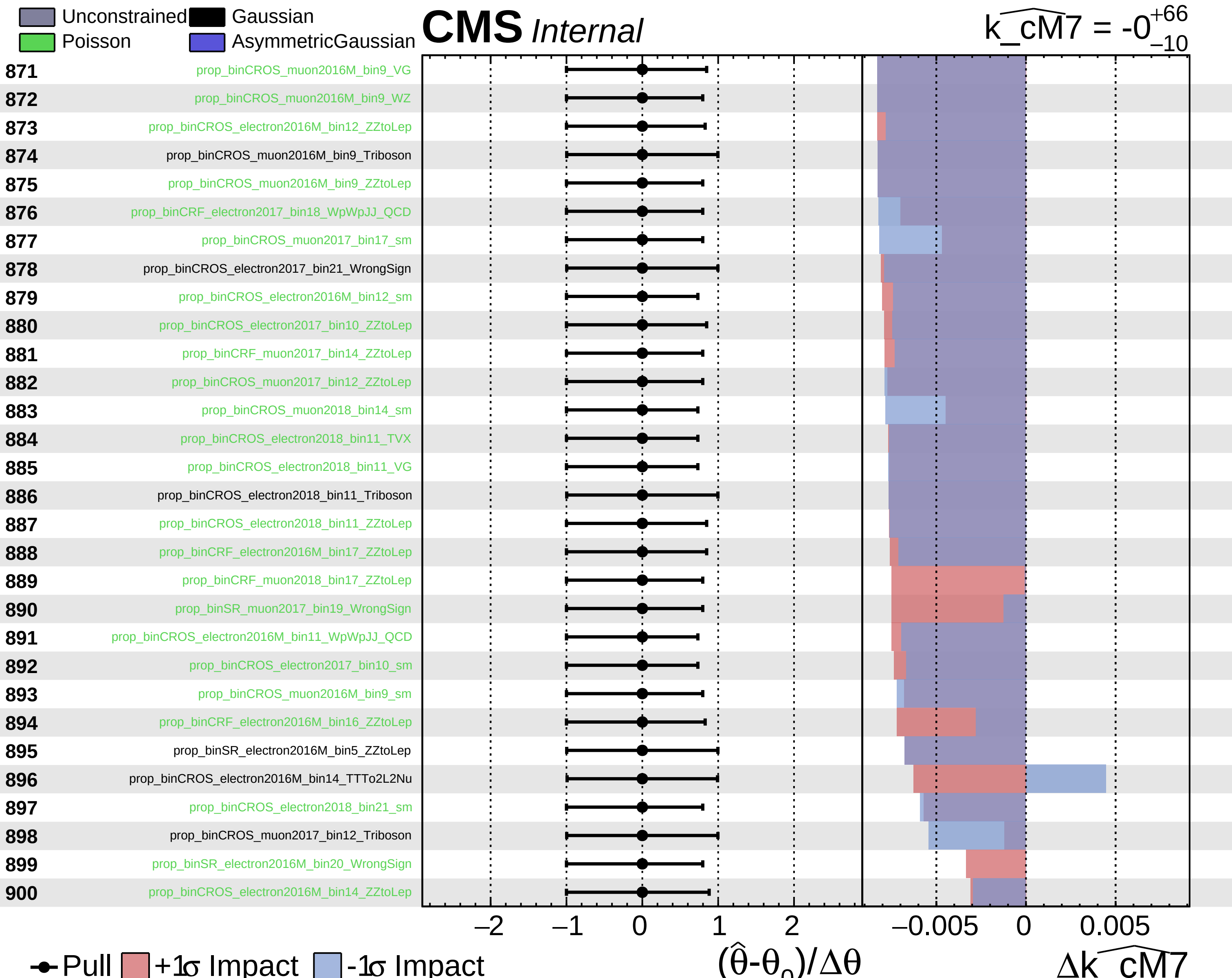
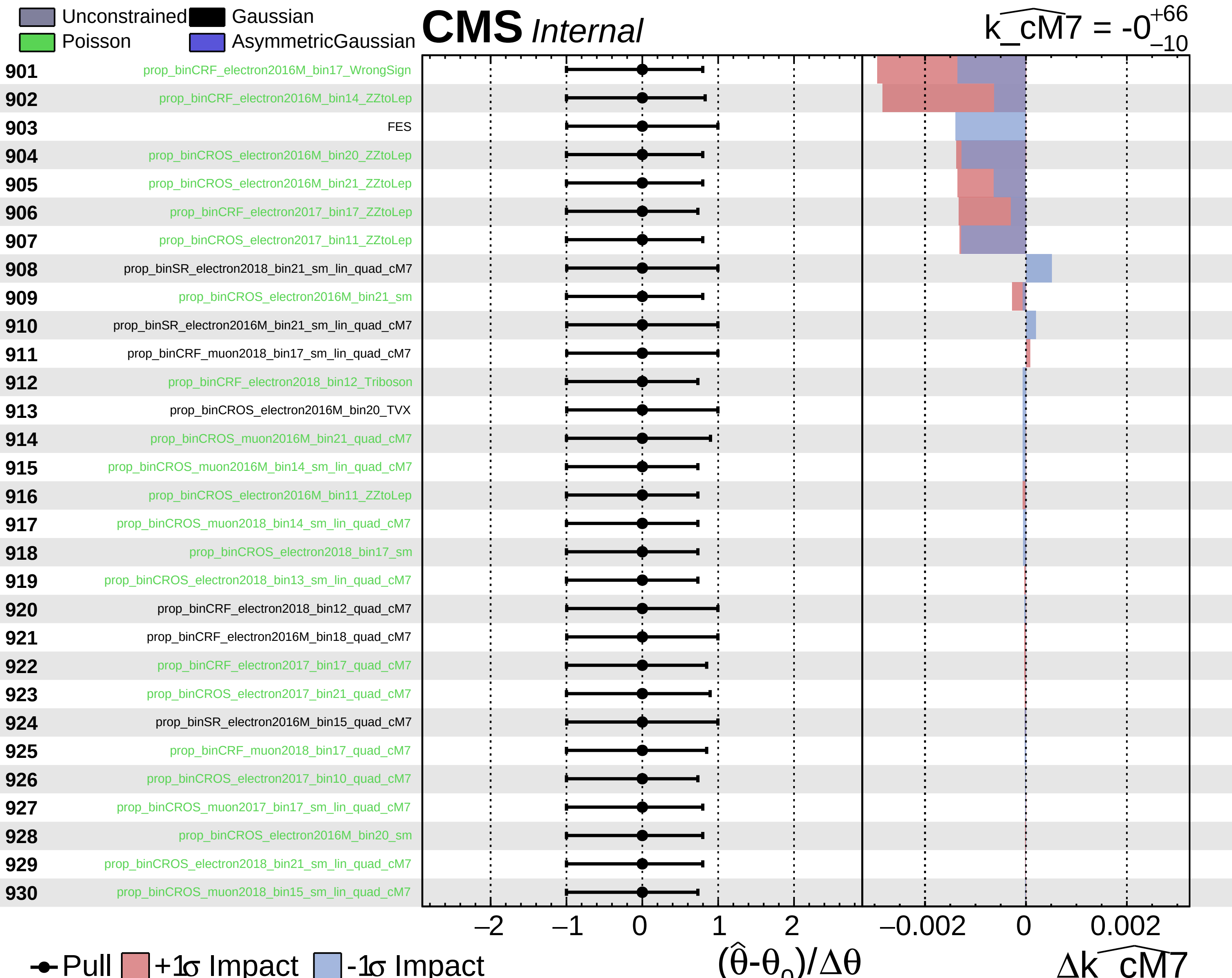
812 prop_binCRF_electron2018_bin12_WpWpJJ_QCD
813 prop_binCROS_muon2018_bin14_TVX
814 prop_binSR_electron2017_bin1
815 prop_binSR_electron2016M_bin5_WpWpJJ_QCD
816 prop_binCROS_electron2018_bin13_ZZtoLep
817 prop_binSR_electron2016M_bin5_TVX
818 prop_binCRF_electron2016M_bin14_WpWpJJ_QCD
819 prop_binCROS_electron2017_bin10_Triboson
820 prop_binCROS_electron2017_bin11_TVX
821 prop_binCRF_muon2017_bin11_ZZtoLep
822 prop_binCROS_electron2018_bin13_Triboson
823 prop_binCROS_muon2017_bin16_ZZtoLep
824 prop_binCROS_electron2018_bin11_TTTo2L2Nu
825 prop_binCRF_muon2016M_bin16_WpWpJJ_QCD
826 prop_binCRF_muon2016M_bin16_Fake
827 prop_binCROS_muon2016M_bin21_TVX
828 prop_binCROS_muon2018_bin12_TVX
829 prop_binCROS_muon2017_bin13_ZZtoLep
830 prop_binCROS_muon2016M_bin7_TVX
831 prop_binCROS_muon2016M_bin7_WZ
832 prop_binCROS_muon2016M_bin7_VG
833 prop_binCROS_electron2017_bin13_TVX
834 prop_binCROS_electron2018_bin17_VG
835 prop_binCROS_muon2016M_bin7_ZZtoLep
836 prop_binCROS_electron2018_bin17_Triboson
837 prop_binCROS_electron2018_bin17_TVX
838 prop_binCROS_electron2018_bin17_ZZtoLep
839 prop_binCROS_electron2016M_bin13_sm
840 prop_binCROS_muon2016M_bin7_Triboson
−2
−1
0
1
2
−0.01
0
0.01
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cM7



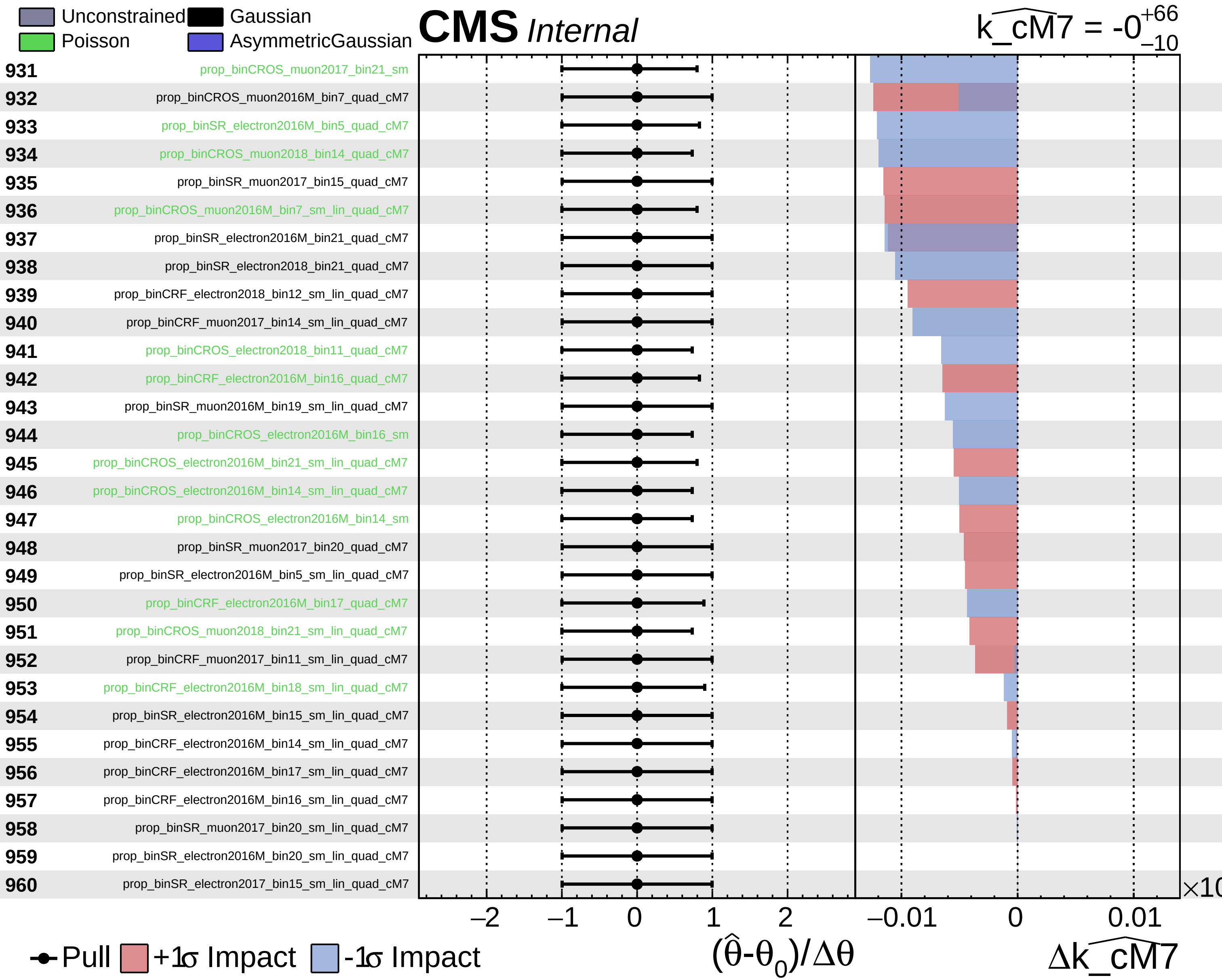
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM7 = -0 +66 −10
841 prop_binCRF_muon2018_bin17_Triboson
842 prop_binCRF_electron2016M_bin17_Triboson
843 prop_binCROS_electron2016M_bin12_VG
844 prop_binSR_electron2017_bin20_WrongSign
845 prop_binCROS_electron2017_bin10_TVX
846 prop_binCROS_muon2018_bin16_ZZtoLep
847 prop_binCROS_electron2017_bin16_ZZtoLep
848 prop_binSR_electron2016M_bin21_ZZtoLep
849 prop_binCROS_electron2016M_bin12_Triboson
850 prop_binCROS_electron2016M_bin12_TVX
851 prop_binCROS_muon2018_bin15_sm
852 prop_binCROS_muon2016M_bin8_ZZtoLep
853 prop_binCROS_muon2016M_bin14_sm
854 prop_binCROS_muon2016M_bin13_sm
855 prop_binCROS_muon2016M_bin9_TTTo2L2Nu
856 prop_binCRF_muon2017_bin14_Triboson
857 prop_binSR_electron2016M_bin21_WrongSign
858 prop_binCROS_muon2017_bin17_ZZtoLep
859 prop_binCROS_muon2016M_bin7_sm
860 prop_binCROS_muon2017_bin21_TVX
861 prop_binSR_electron2016M_bin20_ZZtoLep
862 prop_binCROS_electron2017_bin16_TVX
863 prop_binCRF_muon2016M_bin14_ZZtoLep
864 prop_binSR_electron2017_bin20_ZZtoLep
865 prop_binCRTT_electron2018_bin21
866 prop_binCROS_electron2016M_bin16_ZZtoLep
867 prop_binCROS_muon2016M_bin15_sm
868 prop_binCRF_muon2016M_bin16_Triboson
869 prop_binCROS_electron2018_bin13_sm
870 prop_binCROS_muon2016M_bin9_TVX
−2
−1
0
1
2
−0.01
−0.005
0
0.005
0.01
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cM7



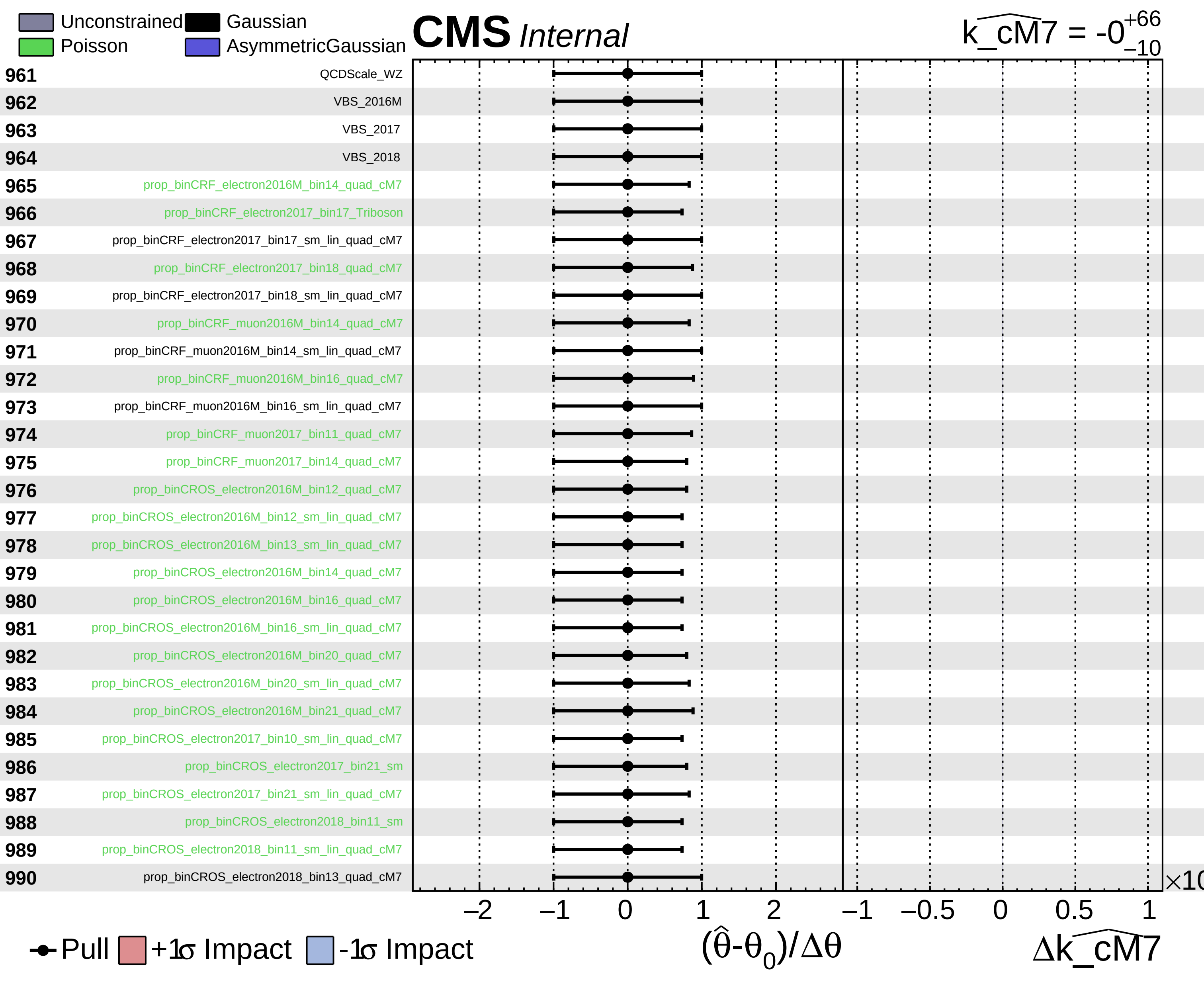
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM7} = -0^{+66}_{-10}$
871 prop_binCROS_muon2016M_bin9_VG
872 prop_binCROS_muon2016M_bin9_WZ
873 prop_binCROS_electron2016M_bin12_ZZtoLep
874 prop_binCROS_muon2016M_bin9_Triboson
875 prop_binCROS_muon2016M_bin9_ZZtoLep
876 prop_binCRF_electron2017_bin18_WpWpJJ_QCD
877 prop_binCROS_muon2017_bin17_sm
878 prop_binCROS_electron2017_bin21_WrongSign
879 prop_binCROS_electron2016M_bin12_sm
880 prop_binCROS_electron2017_bin10_ZZtoLep
881 prop_binCRF_muon2017_bin14_ZZtoLep
882 prop_binCROS_muon2017_bin12_ZZtoLep
883 prop_binCROS_muon2018_bin14_sm
884 prop_binCROS_electron2018_bin11_TVX
885 prop_binCROS_electron2018_bin11_VG
886 prop_binCROS_electron2018_bin11_Triboson
887 prop_binCROS_electron2018_bin11_ZZtoLep
888 prop_binCRF_electron2016M_bin17_ZZtoLep
889 prop_binCRF_muon2018_bin17_ZZtoLep
890 prop_binSR_muon2017_bin19_WrongSign
891 prop_binCROS_electron2016M_bin11_WpWpJJ_QCD
892 prop_binCROS_electron2017_bin10_sm
893 prop_binCROS_muon2016M_bin9_sm
894 prop_binCRF_electron2016M_bin16_ZZtoLep
895 prop_binSR_electron2016M_bin5_ZZtoLep
896 prop_binCROS_electron2016M_bin14_TTTo2L2Nu
897 prop_binCROS_electron2018_bin21_sm
898 prop_binCROS_muon2017_bin12_Triboson
899 prop_binSR_electron2016M_bin20_WrongSign
900 prop_binCROS_electron2016M_bin14_ZZtoLep
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM7}$



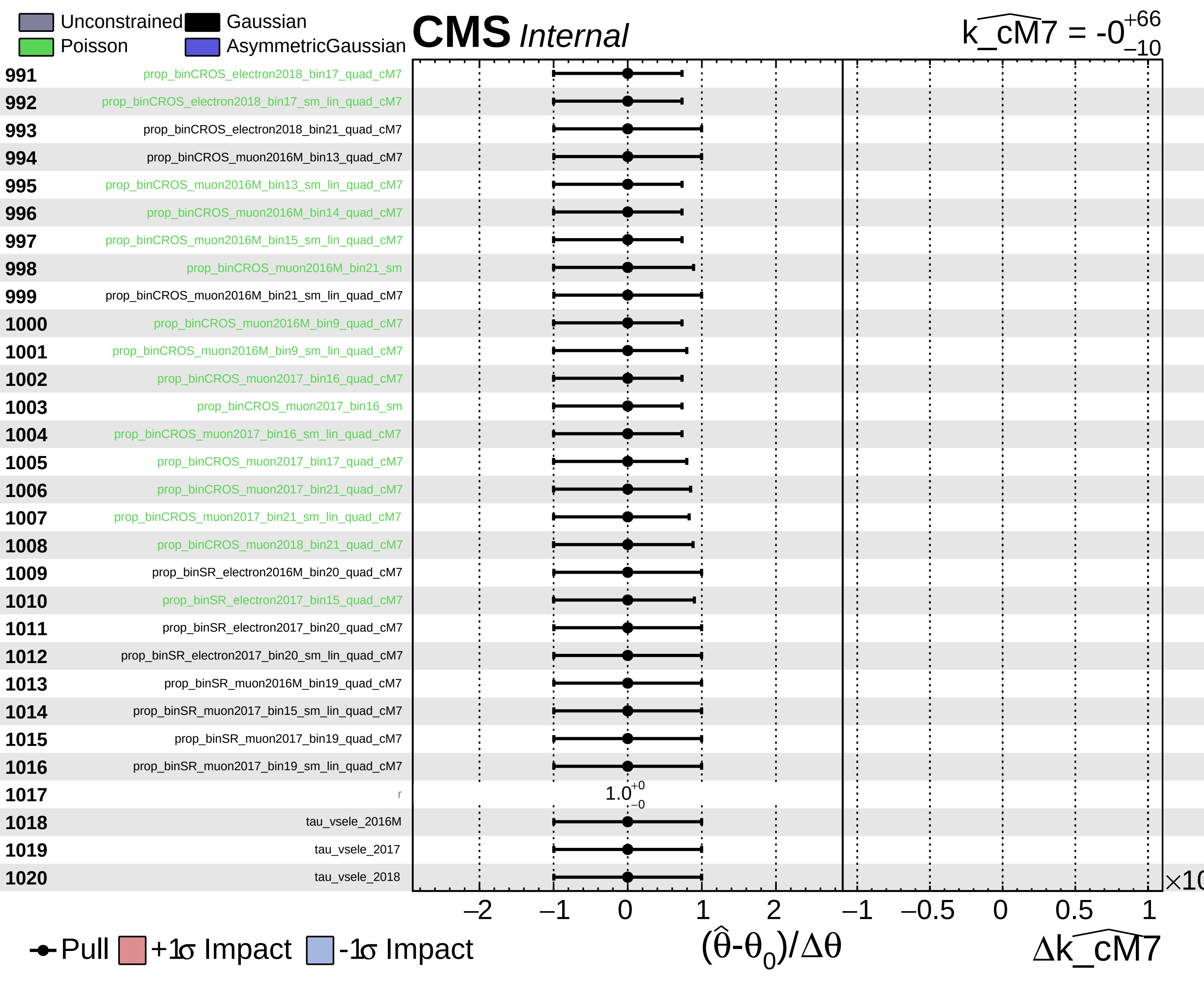
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM7 = -0 +66 −10
901 prop_binCRF_electron2016M_bin17_WrongSign
902 prop_binCRF_electron2016M_bin14_ZZtoLep
903 FES
904 prop_binCROS_electron2016M_bin20_ZZtoLep
905 prop_binCROS_electron2016M_bin21_ZZtoLep
906 prop_binCRF_electron2017_bin17_ZZtoLep
907 prop_binCROS_electron2017_bin11_ZZtoLep
908 prop_binSR_electron2018_bin21_sm_lin_quad_cM7
909 prop_binCROS_electron2016M_bin21_sm
910 prop_binSR_electron2016M_bin21_sm_lin_quad_cM7
911 prop_binCRF_muon2018_bin17_sm_lin_quad_cM7
912 prop_binCRF_electron2018_bin12_Triboson
913 prop_binCROS_electron2016M_bin20_TVX
914 prop_binCROS_muon2016M_bin21_quad_cM7
915 prop_binCROS_muon2016M_bin14_sm_lin_quad_cM7
916 prop_binCROS_electron2016M_bin11_ZZtoLep
917 prop_binCROS_muon2018_bin14_sm_lin_quad_cM7
918 prop_binCROS_electron2018_bin17_sm
919 prop_binCROS_electron2018_bin13_sm_lin_quad_cM7
920 prop_binCRF_electron2018_bin12_quad_cM7
921 prop_binCRF_electron2016M_bin18_quad_cM7
922 prop_binCRF_electron2017_bin17_quad_cM7
923 prop_binCROS_electron2017_bin21_quad_cM7
924 prop_binSR_electron2016M_bin15_quad_cM7
925 prop_binCRF_muon2018_bin17_quad_cM7
926 prop_binCROS_electron2017_bin10_quad_cM7
927 prop_binCROS_muon2017_bin17_sm_lin_quad_cM7
928 prop_binCROS_electron2016M_bin20_sm
929 prop_binCROS_electron2018_bin21_sm_lin_quad_cM7
930 prop_binCROS_muon2018_bin15_sm_lin_quad_cM7
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cM7



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM7 = -0 +66 −10
931 prop_binCROS_muon2017_bin21_sm
932 prop_binCROS_muon2016M_bin7_quad_cM7
933 prop_binSR_electron2016M_bin5_quad_cM7
934 prop_binCROS_muon2018_bin14_quad_cM7
935 prop_binSR_muon2017_bin15_quad_cM7
936 prop_binCROS_muon2016M_bin7_sm_lin_quad_cM7
937 prop_binSR_electron2016M_bin21_quad_cM7
938 prop_binSR_electron2018_bin21_quad_cM7
939 prop_binCRF_electron2018_bin12_sm_lin_quad_cM7
940 prop_binCRF_muon2017_bin14_sm_lin_quad_cM7
941 prop_binCROS_electron2018_bin11_quad_cM7
942 prop_binCRF_electron2016M_bin16_quad_cM7
943 prop_binSR_muon2016M_bin19_sm_lin_quad_cM7
944 prop_binCROS_electron2016M_bin16_sm
945 prop_binCROS_electron2016M_bin21_sm_lin_quad_cM7
946 prop_binCROS_electron2016M_bin14_sm_lin_quad_cM7
947 prop_binCROS_electron2016M_bin14_sm
948 prop_binSR_muon2017_bin20_quad_cM7
949 prop_binSR_electron2016M_bin5_sm_lin_quad_cM7
950 prop_binCRF_electron2016M_bin17_quad_cM7
951 prop_binCROS_muon2018_bin21_sm_lin_quad_cM7
952 prop_binCRF_muon2017_bin11_sm_lin_quad_cM7
953 prop_binCRF_electron2016M_bin18_sm_lin_quad_cM7
954 prop_binSR_electron2016M_bin15_sm_lin_quad_cM7
955 prop_binCRF_electron2016M_bin14_sm_lin_quad_cM7
956 prop_binCRF_electron2016M_bin17_sm_lin_quad_cM7
957 prop_binCRF_electron2016M_bin16_sm_lin_quad_cM7
958 prop_binSR_muon2017_bin20_sm_lin_quad_cM7
959 prop_binSR_electron2016M_bin20_sm_lin_quad_cM7
960 prop_binSR_electron2017_bin15_sm_lin_quad_cM7
−2 −1 0 1 2
−0.01 0 0.01
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cM7



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM7 = -0 +66 −10
961 QCDScale_WZ
962 VBS_2016M
963 VBS_2017
964 VBS_2018
965 prop_binCRF_electron2016M_bin14_quad_cM7
966 prop_binCRF_electron2017_bin17_Triboson
967 prop_binCRF_electron2017_bin17_sm_lin_quad_cM7
968 prop_binCRF_electron2017_bin18_quad_cM7
969 prop_binCRF_electron2017_bin18_sm_lin_quad_cM7
970 prop_binCRF_muon2016M_bin14_quad_cM7
971 prop_binCRF_muon2016M_bin14_sm_lin_quad_cM7
972 prop_binCRF_muon2016M_bin16_quad_cM7
973 prop_binCRF_muon2016M_bin16_sm_lin_quad_cM7
974 prop_binCRF_muon2017_bin11_quad_cM7
975 prop_binCRF_muon2017_bin14_quad_cM7
976 prop_binCROS_electron2016M_bin12_quad_cM7
977 prop_binCROS_electron2016M_bin12_sm_lin_quad_cM7
978 prop_binCROS_electron2016M_bin13_sm_lin_quad_cM7
979 prop_binCROS_electron2016M_bin14_quad_cM7
980 prop_binCROS_electron2016M_bin16_quad_cM7
981 prop_binCROS_electron2016M_bin16_sm_lin_quad_cM7
982 prop_binCROS_electron2016M_bin20_quad_cM7
983 prop_binCROS_electron2016M_bin20_sm_lin_quad_cM7
984 prop_binCROS_electron2016M_bin21_quad_cM7
985 prop_binCROS_electron2017_bin10_sm_lin_quad_cM7
986 prop_binCROS_electron2017_bin21_sm
987 prop_binCROS_electron2017_bin21_sm_lin_quad_cM7
988 prop_binCROS_electron2018_bin11_sm
989 prop_binCROS_electron2018_bin11_sm_lin_quad_cM7
990 prop_binCROS_electron2018_bin13_quad_cM7
−2 −1 0 1 2
−1 −0.5 0 0.5 1
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cM7



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM7 = -0 +66 −10
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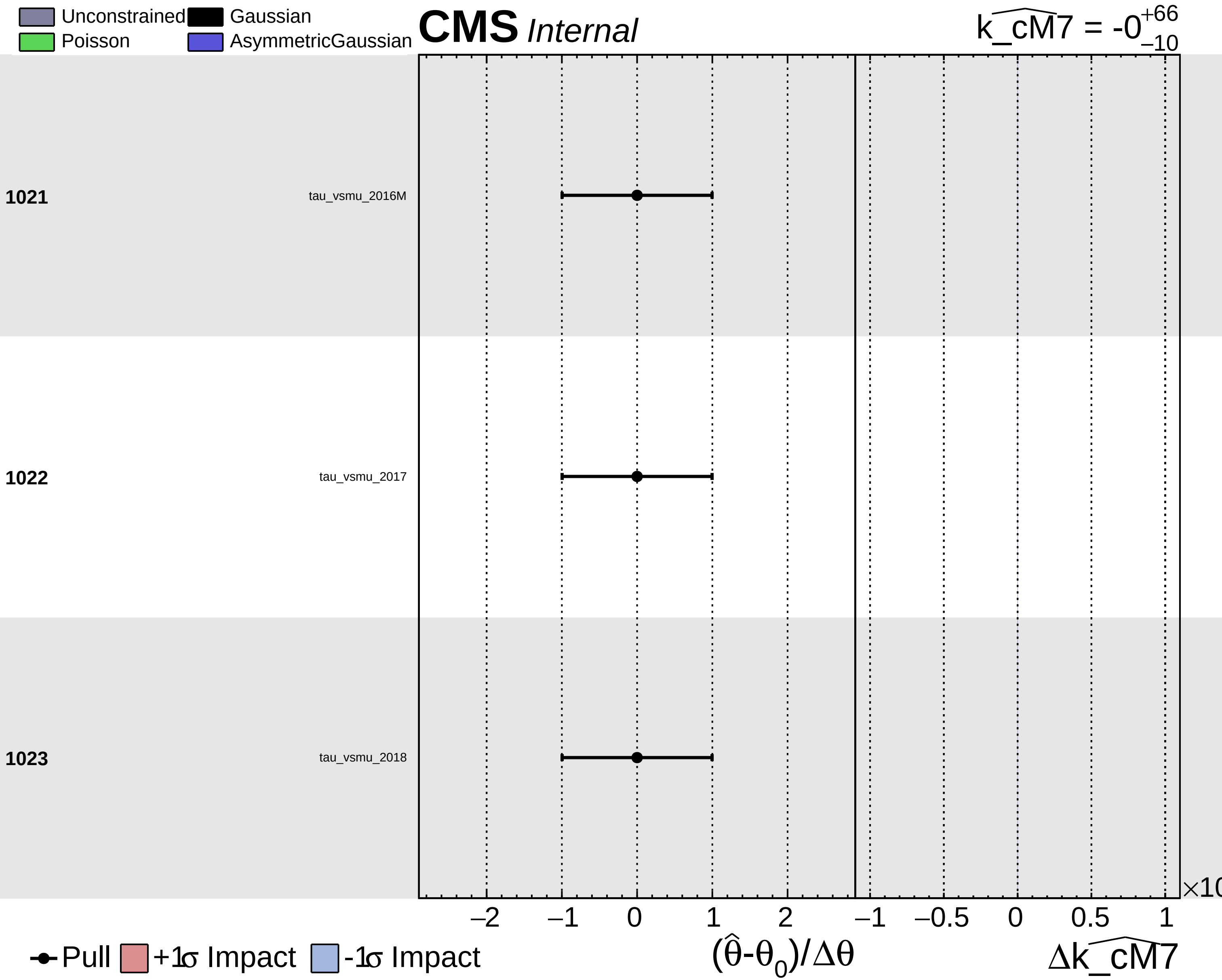
992 prop_binCROS_electron2018_bin17_sm_lin_quad_cM7
993 prop_binCROS_electron2018_bin21_quad_cM7
994 prop_binCROS_muon2016M_bin13_quad_cM7
995 prop_binCROS_muon2016M_bin13_sm_lin_quad_cM7
996 prop_binCROS_muon2016M_bin14_quad_cM7
997 prop_binCROS_muon2016M_bin15_sm_lin_quad_cM7
998 prop_binCROS_muon2016M_bin21_sm
999 prop_binCROS_muon2016M_bin21_sm_lin_quad_cM7
1000 prop_binCROS_muon2016M_bin9_quad_cM7
1001 prop_binCROS_muon2016M_bin9_sm_lin_quad_cM7
1002 prop_binCROS_muon2017_bin16_quad_cM7
1003 prop_binCROS_muon2017_bin16_sm
1004 prop_binCROS_muon2017_bin16_sm_lin_quad_cM7
1005 prop_binCROS_muon2017_bin17_quad_cM7
1006 prop_binCROS_muon2017_bin21_quad_cM7
1007 prop_binCROS_muon2017_bin21_sm_lin_quad_cM7
1008 prop_binCROS_muon2018_bin21_quad_cM7
1009 prop_binSR_electron2016M_bin20_quad_cM7
1010 prop_binSR_electron2017_bin15_quad_cM7
1011 prop_binSR_electron2017_bin20_quad_cM7
1012 prop_binSR_electron2017_bin20_sm_lin_quad_cM7
1013 prop_binSR_muon2016M_bin19_quad_cM7
1014 prop_binSR_muon2017_bin15_sm_lin_quad_cM7
1015 prop_binSR_muon2017_bin19_quad_cM7
1016 prop_binSR_muon2017_bin19_sm_lin_quad_cM7
1017 r
1.0 +0 −0
1018 tau_vsele_2016M
1019 tau_vsele_2017
1020 tau_vsele_2018
−2
−1
0
1
2
−1
−0.5
0
0.5
1
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cM7



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM7 = -0$^{+66}_{-10}$
1021
tau_vsmu_2016M
1022
tau_vsmu_2017
1023
tau_vsmu_2018
×10
−2 −1 0 1 2
−1 −0.5 0 0.5 1
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cM7